Factor Dipping

Yinfeng Zeng*

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Abstract

Standard principal-component (PC) factor estimation relies on a spectral gap that macroeconomic and financial panels routinely lack. This weak-factor problem, combined with the linear pre-processing transformations implicit in applied work, generates PC uncertainty that standard asymptotic intervals severely understate. We propose factor dipping: re-estimate the factors from bounded, invertible transformations of the data, then invert the transformation. Under the standard specification, the dipped and original estimates span the same space up to $O_p(n^{-2})$ deviations, where n is the diverging spectral gap. We turn this invariance into a specification test with a derived limiting distribution, and into a representation-sensitivity band that quantifies the misspecification-induced uncertainty weak factors inject. Applications to Stock–Watson factors, factor-augmented VAR impulse responses, three-pass risk premia, and quantile factor models show that standard intervals can mislead and downstream structural conclusions can be overturned. We recommend factor dipping as a routine diagnostic before any PC-based inference.

JEL classification: C38, C55, C12.

Keywords: factor models; principal components analysis; weak factors; specification testing; model uncertainty; data transformations.

1 Introduction

Factor models are a workhorse of empirical macroeconomics and finance studies and the principal-component (PC) analysis remains the default first pass for factor estimation in

*Corresponding author. Faculty of Economics, University of Cambridge, Sidgwick Avenue, Cambridge CB3 9DD, United Kingdom. Email: yz741@cam.ac.uk.

applied work (for a recent survey, see Bai and Wang (2016)). The asymptotic theory for PC estimation of factor models is well developed. An early seminal contribution is Bai (2003), which supplies the inferential theory and asymptotic confidence intervals for the factors and their loadings. Much of the work that followed adopts Bai (2003)'s assumption of strong factors which implies that the top r eigenvalues of the sample covariance matrix that correspond to r factors are diverging at rate N, where N is the cross-sectional dimension of the data. This divergence opens a spectral gap between the r-th and $r+1$-th largest eigenvalues of the sample covariance matrix, $n \equiv \lambda_r/\lambda_{r+1}$. However, typical datasets do not show a clear gap in the spectrum of the sample covariance. The scree plot of the Stock–Watson–McCracken–Ng panel (McCracken and Ng, 2016) in Figure 1 is representative. The eigenvalues decline smoothly, and the eigenvalue of the weakest selected (the eighth) "factor" is barely distinguishable from the idiosyncratic bulk.

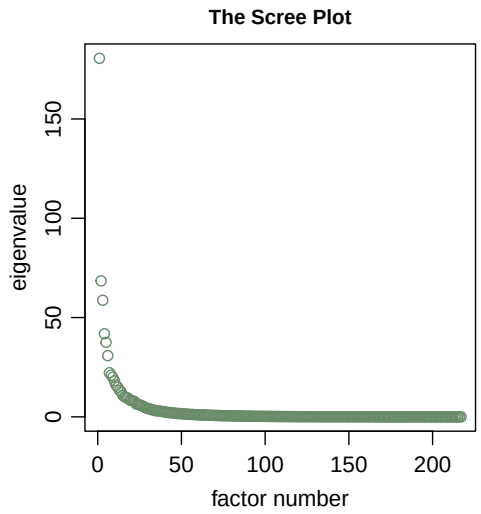


Figure 1: The scree plot for the Stock-Watson macroeconomic data.

Recent work on weak factors, briefly surveyed in Section 1.2, recognizes the problem. Once factors and loadings are allowed to be weaker, the spectral gap still diverges, but at a slower rate N^α with $\alpha \in (0, 1]$. Bai and Ng (2023) and Fan et al. (2024) extend valid inference into this semi-strong regime. However, existing measures can be unreliable for weaker factors.

When the gap is narrow, the point estimate can depend on choices an applied researcher would consider innocuous, and standard confidence intervals can understate the true PC uncertainty by a wide margin in finite samples (for example, see Figure 2, Table 4 and Figure 3 in Section 7).

A concrete illustration of a problem caused by the absence of clear separation between factor and idiosyncratic eigenvalues is the exercise of Stock and Watson (2016), who revisit Sims's question of whether a small vector autoregression (VAR) approximates the dynamic factor model by asking whether a handful of observable variables span the same space as the eight factors extracted by PC from a large macroeconomic panel. They compute canonical correlations between three standard VAR variable sets and the estimated factors (their Table 5, reproduced in Table 1); several fall well below one, and they conclude that the large-panel factors carry information the small VARs miss. They are scrupulous about the caveat: "Because the factors are estimated, the sample canonical correlations will be less than one even if in population they equal one, and no measure of sampling variability is provided." The substantive conclusion therefore rests on a quantity whose finite-sample behaviour is uncharacterized—and, as we show, whose behaviour a marginal spectral gap renders fragile in a way no reported standard error would reveal.

Table 1: Approximating the eight-factor DFM by an eight-variable VAR - Panel B of Table 5 in Stock and Watson (2016)

	Canonical Correlation							
variables and factors	1	2	3	4	5	6	7	8
VAR - B	0.97	0.95	0.89	0.83	0.61	0.43	0.26	0.10
VAR - C	0.98	0.93	0.90	0.87	0.79	0.78	0.57	0.41
VAR - O	0.98	0.96	0.88	0.84	0.72	0.39	0.18	0.02

To make this fragility visible, testable, and quantifiable, we propose a novel procedure named *factor dipping*. Consider the standard approximate factor model

$$X = F\Lambda' + e, \tag{1.1}$$

where X is $T \times N$, F is $T \times r$, Λ is $N \times r$, and e is idiosyncratic, and where the number of factors r is taken as given—the number the researcher has already chosen to work with. Let A be any bounded and invertible linear transformation matrix with bounded inverse. Extract the PC factors $\hat{F}_A$ for the transformed factors (AF) from the transformed panel AX and invert to form the *dipped estimate* $\tilde{F} = A^{-1}\hat{F}_A$. Informally, our proposed procedure is like dipping a pancake (the factor) into honey (applying A) and lifting it out (A^{-1})—it tastes different afterwards ($\hat{F} \neq \tilde{F}$) but remains the same pancake (both estimate the same F). Under the standard specification, $\hat{F}$ and $\tilde{F}$ are valid alternatives to one another.

Applied to the Stock and Watson (2016) exercise, mild symmetric Toeplitz (Kac–

Table 2: Canonical correlations between original PC factor estimates (Stock and Watson (2016)) and dipped factor estimates under four symmetric Toeplitz transformations $\mathbf{A}_T(a)$.

Transformation	Canonical Correlation by Factor							
	1	2	3	4	5	6	7	8
$a = 0.5$	1.000	1.000	1.000	1.000	1.000	1.000	0.994	0.989
$a = -0.5$	1.000	1.000	1.000	1.000	0.999	0.990	0.982	0.492
$a = 0.6$	1.000	1.000	1.000	1.000	1.000	1.000	0.991	0.984
$a = -0.6$	1.000	1.000	1.000	1.000	0.995	0.983	0.897	0.382

Murdock–Szegő) transformations $\mathbf{A}_T(a)$ with $a = \pm 0.5, \pm 0.6$—whose spectral norms, roughly 3 to 4, are dominated by $\|F\Lambda'\| \approx 236.78$, so that no factor is removed and no idiosyncratic variation is overly amplified—leave the first six canonical correlations at unity but collapse the eighth to 0.49 at $a = -0.5$ and 0.38 at $a = -0.6$ (Table 2), beneath the 0.5 threshold on which the original reading turns. As we prove, deviations of this magnitude are impossible in the presence of a clean spectral gap. Their appearance is direct evidence that the weakest factors are estimated imprecisely to sustain the original inference, so the case against small VARs can be less conclusive than it appears.

We prove formally an invariance theorem. Under the standard specification—allowing weaker loadings with $\|\Lambda'\Lambda\| = O_P(N^\alpha)$, $\alpha \in (0, 1]$—the dipped and original estimates span the same space, their canonical correlations departing from unity only at rate $O_P(n^{-2})$ (Theorem 2.1).[1] When the gap is wide, the transformation is essentially undone by A^{-1} and $\hat{F}, \tilde{F}$ coincide; when it is narrow, A mixes idiosyncratic directions into the weakest factors and undoing it recovers a materially different estimate. Strong factors thus render the representation immaterial, weak factors make it decisive—and this breakdown yields two deliverables.

The first is a rigorous test of the standard specification, which the literature has lacked. The eye-judgment of the scree plot is subjective and scale-dependent, and existing tests target factor strengths or counts (Onatski, 2009, 2010; Ahn and Horenstein, 2013; Bailey et al., 2021) rather than the specification itself. Our statistic is constructed based on the deviation $S = r - \mathrm{tr}[P_{\hat{F}} P_{\tilde{F}}]$ (Sections 3–3.2). Applied sequentially to successively larger subsets of the retained factors, the same statistic identifies the number of *stable factors* $r_s \leq r$, a stability-based notion of dimensionality complementing the variance- and penalty-based criteria (Bai and Ng, 2002; Ahn and Horenstein, 2013). The logic is that of a Hausman test (Hausman, 1978), where $\hat{F}$ and $\tilde{F}$ agree under correct specification and diverge otherwise.

[1] The second-order agreement echoes index-number theory, where superlative indices coincide to second order regardless of the admissible formula (Diewert, 1976).

The transform–estimate–revert structure mirrors PANIC (Bai and Ng, 2004), the difference being that dipping deploys a whole family of transformations to test a property.

The second deliverable is a representation-sensitivity band. The envelope of dipped estimates over the library quantifies the misspecification-induced uncertainty that weak factors inject, and we propagate it through downstream analyses. The band has a design-based reading (Abadie et al., 2020; Rambachan and Roth, 2025), where each transformation is an intervention on the data representation, the ensemble of dipped estimates is the design distribution, and the invariance theorem says that its treatment effect is zero under correct specification—a design distribution that, unlike in causal settings, is directly observed. Note that this is an analogy, not an equivalence. The library is analyst-chosen, so the envelope is a sensitivity band rather than a coverage statement.

Why should transforming the data reveal anything at all? Because there is no untransformed data to begin with. By the time X reaches the analyst it has passed through a long chain of essentially linear operations: the choice of units and functional form, real-versus-nominal conversion (deflation is a diagonal transform), seasonal adjustment (a linear filter), base-year rebasing, temporal aggregation, the sampling frequency, and the vintage that was downloaded. The FRED–MD codebook (McCracken and Ng, 2016) makes the point starkly, assigning each of 128 series a transformation code before any eigenvector is computed, so that the most widely used macroeconomic panel arrives with a specific composite A already applied. The identity map "$A = I$" is therefore not the absence of a design but one specific, historically contingent representation that convention happened to supply. A researcher who "does not transform the data" has merely accepted the representation that was assigned. It follows that the sensitivity of PC estimates to A is not an artifact introduced by dipping but a latent property of the data and the estimand that any single representation conceals. When the factors are strong, our theorem guarantees this sensitivity is negligible; when they are weak it is first order, and reporting a single representation can misstate what the data support. A body of work has documented piecemeal that particular pre-PC transformations shift the extracted factors—standardization (Greenaway-McGrevy et al., 2012), pre-screening (Boivin and Ng, 2006), HP and band-pass filtering (Hodrick and Prescott, 1997; Coulombe et al., 2021), even differencing a single series (Hamilton et al., 2024). The transformation A is the common abstraction of all of these—detrending is a projection, deflation a diagonal map, filters Toeplitz operators, portfolio formation a linear combination—and factor dipping treats the whole catalogue as one general object, asking with a theorem to calibrate the answer whether the choice among its members matters for the estimate at hand.[2]

[2]First-differencing lies outside our admissible class; it is the boundary case, and extending the diagnostic

Factor dipping delivers a warning: the number of factors a researcher retains carries a consequence for the stability of PC inference that conventional standard errors do not reveal. Retaining more, weaker factors explains more of the panel's variation—a genuine benefit—but yields a less clean spectral gap, injecting a misspecification-induced source of PC uncertainty that the standard Bai (2003) intervals leave out. Factor dipping makes this alternative, otherwise-hidden source visible. One who values stability may trade a little explained variation for it, moving to a more conservative count whose PC space is robust across representations; one who prefers to keep the weak factors for information may bear the misspecification, but should at least do so knowing the uncertainty it induces—an uncertainty that is not safely bounded by the standard measure. In this sense, dipping is a diagnostic to report alongside point estimates. Two boundaries guard against overstatement: dipping recommends no representation, and it cannot repair weak-factor identification. The distributional theory requires the dip to be asymptotically free of the idiosyncratic dependence, so a structured temporal filter can mislead under serially persistent idiosyncratic terms, whereas the random members of our library are free of that structure by construction (Online Appendix).

1.1 Contributions

Our contributions are threefold. (i) We establish an invariance property of the factor structure—the space recovered by principal components is preserved, to order $O_p(n^{-2})$ under any bounded invertible linear transformation—which may be of independent interest[3]. (ii) We provide the direct and rigorous test of the standard factor-model specification, in place of scree-plot inspection. (iii) We introduce a representation-sensitivity band that quantifies, as directly observed design-based uncertainty, the misspecification-induced PC uncertainty a marginal gap creates to each factor. Cutting across these, the analysis carries a practical warning—the validity of the standard specification should be tested rather than assumed, and any PC-based inference should be accompanied by a factor-dipping diagnostic, much as robust standard errors accompany a regression today.

to the level-versus-difference decision, in the spirit of Bai and Ng (2004) and Hamilton et al. (2024), is left for future work.

[3]The result characterizes when principal-subspace estimates are robust to the reweighting and filtering choices long flagged as consequential—from the classical non-invariance of covariance- versus correlation-based PCA (Jolliffe, 2002) to the invariant-subspace perturbation theory underlying modern spectral methods (Davis and Kahan, 1970; Stewart and Sun, 1990; Yu et al., 2015)—and it makes explicit the invariance that several estimators invoke implicitly (Bai and Ng, 2004; Choi, 2012; Fan et al., 2016; Giglio and Xiu, 2021).

1.2 Related literature

PC-based estimators whose inferential theory, like Bai (2003), presupposes the gap include the GLS-type PC (Choi, 2012), projected PCA (Fan et al., 2016), QML dynamic factor models (Doz et al., 2012), supervised and instrumented PCA (Bair et al., 2006; Giglio et al., 2021; Kelly et al., 2019), the three-pass estimator (Giglio and Xiu, 2021), sparse and robust PCA (Johnstone and Lu, 2009; Candès et al., 2011), and matrix/tensor factor models (Wang et al., 2019; Chen et al., 2022). Because dipping acts on the data before the eigen-decomposition, it applies to all of them unchanged, and asks the one question they presuppose but none test: whether the retained structure is separated enough to be representation-robust in finite samples. The same premise increasingly underlies causal and policy evaluation, where principal-component regression and low-rank matrix methods drive counterfactual estimation—robust and PCR-based synthetic controls (Amjad et al., 2018; Agarwal et al., 2021) and the factor-model view of synthetic difference-in-differences (Arkhangelsky et al., 2021)—each extracting latent factors from a donor panel and building the counterfactual on the retained principal components.

The number-of-factors tests and estimators (Onatski, 2009, 2010; Ahn and Horenstein, 2013; Bailey et al., 2021), detection-limit theory (Onatski et al., 2013; Dobriban, 2017; Trapani, 2018), and the random-matrix foundations (Johnstone, 2001; Baik et al., 2005; Paul, 2007) that consider a weak factor a near-critical spike make the gap itself the object. These literature measure the size of the gap; dipping assesses it and measures its consequence for the PC stability of the estimate.

The literature on weak factors recognizes the absence of a clean spectral gap. Allowing a slower divergence rate N^α of the factor eigenvalues, Bai and Ng (2023) and Fan et al. (2024) provide the confidence intervals of factors and loadings. See also Bailey et al. (2021); Freyaldenhoven (2022); Uematsu and Yamagata (2023). In the absence of an asymptotic gap ($\alpha = 0$), Onatski (2012) provides the inferential theory under an assumption of asymptotic gap in eigenvalue increments, $(\lambda_r - \lambda_{t+1})/(\lambda_{r+1} - \lambda_{r+2}) \to \infty$. In this regime, under gaussianity, the estimated factors are asymptotically rotated and contaminated versions of the true factors. A recent contribution of Bykhovskaya et al. (2025) develops inferential theory that works even without such a gap in increments, but only for the factor strengths (ℓ_2 norm of the loadings) under gaussianity and independently and identically distributed idiosyncratic terms. They show that near the phase-transition threshold separating recoverable from non-informative signals, the Gaussian approximations underlying standard eigenvalue-based inference break down.

The design-based reading connects dipping to the robustness of conclusions to defensible but arbitrary analytical choices—specification-curve or multiverse analysis (Simonsohn et al.,

2020; Steegen et al., 2016)—to which dipping adds an invariance theorem that pins down when across-representation variability should be exactly zero. The resulting envelope aligns with the design-based uncertainty (Abadie et al., 2020), a partial-identification inner bound (Manski, 2003), and a distributionally-robust object (Hansen and Sargent, 2001).

1.3 Applications

Because factor dipping acts before the eigen-decomposition, its reach is as wide as that of PC estimation itself, and we develop three applications in which the stakes are clear. We first apply the diagnostic on the quantile factor analysis (Chen et al., 2021). The second is the estimation of risk premia by the three-pass procedure of Giglio and Xiu (2021), where the premium of an observable factor is read off a PC-recovered factor space and inherits its fragility when that space is weakly separated. The third is the factor-augmented VAR analysis, where, taking the Bernanke et al. (2005) specification as given, we show that the structural impulse responses built on that space nonetheless range from representation-robust (the exchange-rate response) to representation-fragile (the sign of the equity response, and the price "puzzle" that motivated the exercise), a per-response credibility check invisible to a confidence band computed at a single specification.

1.4 Outline

The rest of the paper is organized as follows. Section 2 states the framework, proves the invariance theorem, and introduces factor dipping formally. Section 3 derives the asymptotic distribution of the specification statistic in the one-factor case, and Section 3.2 extends it to the multi-factor omnibus test. Section 4 summarizes the Monte Carlo evidence—size, power, robustness to non-Gaussianity, and the near-unit-root boundary—reported in full in the Online Appendix. Section 5 develops the uncertainty-assessment procedures and their interpretation, Section 6 discusses the choice of transformation library, and Section 7 presents the empirical applications. Section 8 concludes. The proofs of the main results are collected in the Appendix; auxiliary computations, the robustness extensions, and the simulation study are in the Online Appendix.

2 Invariant Factor Structure and Factor Dipping

2.1 The general framework

We consider the r-factor model as described in (1.1) together with the linearly transformed factor model:

$$X = F\Lambda' + e \qquad \text{and} \qquad AX = AF\Lambda' + Ae.$$

Here A is a $T \times T$ matrix transforming the data. To identify F and Λ we normalize $F'F/T = I_r$ with $\Lambda'\Lambda$ diagonal in descending order; the PC estimator $\hat{F}$ is $\sqrt{T}$ times the eigenvectors of XX'/T for the top r eigenvalues, and $\hat{\Lambda} = X'\hat{F}/T$. Let $\lambda_1 > \cdots > \lambda_T$ be the eigenvalues of XX' and $\|\cdot\|$ the spectral norm. We make three relatively weak assumptions:

A1 $\|e\|^2 / \lambda_r = o_P(1)$ as $N, T \to \infty$.

A2 $\|e\|^2 / \lambda_{r+1} = O_P(1)$ as $N, T \to \infty$.

A3 $\|A\| = O_P(1)$ and $\|A^{-1}\| = O_P(1)$ as $T \to \infty$.

By the singular-value inequalities for a sum of matrices (Thompson, 1975), **A1** is equivalent to the norm of e being negligible relative to the r-th singular value of $F\Lambda'$—a measure of the weakest factor's strength—so all r factors dominate the idiosyncratic term, without specifying the rate.

Assumption **A2** aids identification: it can fail only if the leading singular directions of e align with F or Λ, making systematic and idiosyncratic variation hard to distinguish. A sufficient condition is that the first and $(r+1)$-th singular values of e are of the same order, which holds for a very large class of random matrices whose singular values pack into an absolutely continuous bulk.

Assumption **A3** ensures that neither A nor A^{-1} is so dominant as to remove a factor or inflate the idiosyncratic variation, keeping the factor model intact after transformation. It restricts only the researcher's choice of A, not the data, and can always be met; first-differencing does not satisfy it.

2.2 Factor dipping

We propose *factor dipping*—a simple, computationally inexpensive, and broadly applicable procedure that generates alternative factor (loading) estimates to the standard Principal Component (PC) estimates. The core idea is elementary—the validity of the factor structure is preserved under linear transformation, given the standard specification.

Let A be any bounded invertible linear transformation applied to the $T \times N$ data matrix X. Extract and estimate PC factors $\hat{F}_A$ (PC estimator for AF) from AX and invert to recover

$$\tilde{F} = A^{-1}\hat{F}_A, \tag{2.1}$$

an alternative to $\hat{F}$. This is motivated by PANIC (Bai and Ng, 2004), which applies PC to first-differenced data and cumulates the estimated factors back to levels; the transform–estimate–untransform structure is the same, though differencing itself is inadmissible under A3. Our assumptions on A are far weaker, admitting a wide range of transformations, and we characterize precisely when and to what degree the factor *estimates*—not merely their consistency—are preserved, and make the deviations testable.

Under the standard specification (allowing for weaker loadings such that $\|\Lambda'\Lambda\| = O_P(N^\alpha)$ with $\alpha \in (0, 1]$), $\hat{F}$ and $\tilde{F}$ estimate the same object: they span the same factor space and canonical correlations between them deviate from unity only at rate $O_P(n^{-2})$, where $n = \lambda_r/\lambda_{r+1}$, the spectral gap of the eigenvalues of XX' (Theorem 2.1). We prove this (Theorem 2.1) by a perturbation argument sketched in Section 2 and completed in the Online Appendix. When weak factors are present the deviations can inflate: a linear transformation changes how the true factors and idiosyncratic terms map into the eigenvectors of the sample covariance, and when the separation between weak factors and noise is not sharp, the transformation mixes idiosyncratic directions into the weaker factors, so undoing it no longer recovers the original estimate.

2.3 Invariant factor structure

We now show that the factor structure is invariant to bounded linear transformations: under the standard specification the dipped estimates span the same space as the original PC estimates, with deviation of order $O_P(n^{-2})$. The argument is a second-order eigen-projection perturbation. We give the setup, notation, and theorem here, and defer the algebraic manipulations to Appendix A.

Write $\mathcal{F} = F(\Lambda'\Lambda)^{1/2} + e\Lambda(\Lambda'\Lambda)^{-1/2}$, so that $XX' = \mathcal{F}\mathcal{F}' + eM_\Lambda e'$ splits into a factor part and an idiosyncratic part; let P_0 project onto the factor space $\text{span}(\mathcal{F})$, $M_0 = I - P_0$, and $W = A'A$. The normalized operator $\Phi = \lambda_r^{-1}XX' = \Phi_0 + \Phi_1/n$, with $\Phi_0 = \lambda_r^{-1}\mathcal{F}\mathcal{F}'$ and $\Phi_1 = \lambda_{r+1}^{-1}eM_\Lambda e'$, is an $O_P(n^{-1})$ perturbation of the factor operator Φ_0, so the eigen-projection formulae of Kato (1980, pp. 75–77) apply. The mechanism is then transparent. To first order the transformation is undone by A^{-1} and the two projections agree, so their traces cancel and $S \equiv r - \text{tr}[P_{\hat{F}}P_{\tilde{F}}] = O_P(n^{-2})$: the dipped and original spaces coincide to order n^{-2}. At second order the annihilation identity $M_0A^{-1}P_{Aj} = 0$—valid because $A^{-1}P_{Aj}$ maps

into span($\mathcal{F}$), which M_0 kills—collapses the cross terms and leaves a single quadratic form in the contrast operator $D_A \equiv A'\Phi_{A0}^{+}A - \Phi_0^{+}$ (a subscript A denotes a quantity formed from AX). Appendix A carries out both expansions in full, including the rank-r group-projector form for clustered factor eigenvalues.

Theorem 2.1. *Consider an r-factor model $X = F\Lambda' + e$ and a $T \times T$ matrix A that linearly transforms the model to $AX = AF\Lambda' + Ae$. Let $\lambda_1 > \lambda_2 > \cdots > \lambda_T$ be the eigenvalues of XX', $n \equiv \lambda_r/\lambda_{r+1}$, and $S \equiv r - \mathrm{tr}[P_{\hat{F}}P_{\tilde{F}}]$, where $P_{\hat{F}}$ and $P_{\tilde{F}}$ are the projections onto the column spaces of the PC estimate $\hat{F}$ and the dipped estimate $\tilde{F} = A^{-1}\hat{F}_A$. Under Assumptions A1–A3,*

$$S = r - \mathrm{tr}[P_{\hat{F}}P_{\tilde{F}}] = O_P(n^{-2}), \qquad n^2 S = \|P_0 D_A \Phi_1 M_0\|_F^2 + O_P(n^{-1}),$$

where $\|\cdot\|_F$ is the Frobenius norm, $\Phi_1 = \lambda_{r+1}^{-1} e M_\Lambda e'$, P_0 ($M_0 = I - P_0$) is the projection onto (orthogonal to) the factor space, and $D_A \equiv A'\Phi_{A0}^{+}A - \Phi_0^{+}$ with Φ_0^{+}, Φ_{A0}^{+} the rank-r Moore–Penrose inverses of Φ_0 and Φ_{A0}.

This invariance property of the factor structure may be of independent interest. The deviation S of $\mathrm{tr}[P_{\hat{F}}P_{\tilde{F}}]$ from r is the statistic of our specification test, whose asymptotic distribution is derived in Section 3.

3 The Specification Test

Despite the scree plot showing how the eigenvalues are separated, using it as evidence for the validity of the standard specification is subjective because it depends on eye judgment and the plot scale. Our test provides a rigorous alternative that assesses the invariance of the factor structure under linear transformations. In spirit it resembles the bootstrap specification test for GARCH models of Cavaliere et al. (2024): a parametric bootstrap generates series from the fitted null and asks whether the observed statistic looks like a typical draw; here, factor dipping generates alternative estimates under the maintained specification and asks whether the observed deviation between dipped and original estimates exceeds what the null admits.

The test exploits the fact that, under the null, any "dipped" factor estimate $\tilde{F}$ spans essentially the same space as the original PC estimate $\hat{F}$ (proven in section 2). Define the orthogonal projections onto the column spaces of $\hat{F}$ and $\tilde{F}$ as $\hat{P}_{\hat{F}}$ and $\tilde{P}_{\tilde{F}}$, respectively. Our test statistic is, as mentioned previously,

$$S = r - \mathrm{tr}(\hat{P}_{\hat{F}}\tilde{P}_{\tilde{F}}). \tag{3.1}$$

The null hypothesis. Fix a working number of factors r and a dip A satisfying Assumption **A3**. The invariance of Theorem 2.1 is driven by Assumption **A1**, which requires the r-th factor to dominate the idiosyncratic spectrum; the test is therefore a test of **A1** at rank r, i.e. of whether the standard PC specification supports valid inference on an r-dimensional factor space. Writing $n = \lambda_r/\lambda_{r+1}$ for the spectral gap, we test

$$\begin{aligned} \mathrm{H}_0 : \ & \frac{\|e\|^2}{\lambda_r} = o_P(1) \quad \big(\text{equivalently } n \xrightarrow{P} \infty\big), \\ \text{against} \quad \mathrm{H}_1 : \ & \frac{\|e\|^2}{\lambda_r} \text{ is bounded away from 0} \quad \big(\text{equivalently } n = O_P(1)\big). \end{aligned} \tag{3.2}$$

so that under H_1 the r-th eigen-direction is not asymptotically separated from the idiosyncratic bulk. (Under Assumption **A2**, $\|e\|^2 \asymp \lambda_{r+1}$, so $\|e\|^2/\lambda_r \to 0$ is equivalent to $n \to \infty$.) Under H_0, Theorem 2.1 gives $S = O_P(n^{-2}) \to 0$ and the scaled statistic has the limiting law of Theorem 3.2 below; under H_1 the trace cancellation that produces the n^{-2} rate fails, S is of larger order, and large values of S lead to rejection (Proposition 3.3).

A rejection licenses the conclusion that PC inference at the working rank r is fragile: the r-dimensional PC space is not robust to the admissible data representations, so the Bai (2003) intervals built on it need not be valid. A rejection is not, by itself, evidence that the true number of factors differs from r: the test certifies the robustness of a working space, it does not count factors. This is what distinguishes dipping from the number-of-factors estimators of Bai and Ng (2002) and the edge/eigenvalue tests of Onatski (2009, 2010), which estimate how many factors are present; dipping instead identifies the largest sub-block whose PC inference is robust, formalized as $\hat{r}_{\mathrm{dip}}$ in Section 3.4.

3.1 Asymptotic Distribution of the Test Statistic

For demonstration and simplicity, we derive the asymptotic distribution of the test statistic for the one factor case ($r = 1$). Consider

$$X = F\Lambda' + e, \qquad X \in \mathbb{R}^{T\times N},$$

where $F \in \mathbb{R}^T$ is the factor, $\Lambda \in \mathbb{R}^N$ is the vector of loadings, and $e \in \mathbb{R}^{T\times N}$ is the idiosyncratic matrix. We maintain Assumptions A1-A3. Additionally, for the distributional results we impose:

B1 $F = \Sigma_F^{1/2}\xi$ with $\xi \sim N(0, I_T)$, independent of e.

B2 The idiosyncratic matrix satisfies $e = \varepsilon\Sigma^{1/2}$, where ε is a $T \times N$ matrix of i.i.d. $N(0,1)$

entries and $\Sigma \in \mathbb{R}^{N\times N}$ is a positive-definite covariance matrix.

B3 The normalization $T^{-1}\operatorname{tr}\Sigma_F = 1$ is imposed for identification of Σ_F.

B4 $T/N \to c \in (0,\infty)$ and $n \gg T^{1/2}$ as $N, T \to \infty$.

Assumption **B4** is stated as a high-level rate on the gap n. It is useful to record what it requires of the factor strengths, and how it can be read off the data. Note that it is needed only for the Gaussian null law of Theorem 3.2. We therefore read the null of Theorem 3.2 as the stable-factor null $n \gg \sqrt{T}$, against the alternative $n = O_P(\sqrt{T})$, which subsumes genuinely weak factors and outright misspecification and against which the test is consistent. For $\alpha_r \le \frac{1}{2}$ invariance and consistency still hold but the Gaussian size is not controlled; there the reported object is the representation-sensitivity envelope of Section 5, which carries no size claim.

Lemma 3.1 (Primitive form of the gap condition)**.** *Adopt the strength parameterization of the Online Appendix, under which the k-th factor eigenvalue of XX'/T satisfies $\lambda_k \asymp_P N^{\alpha_k}$ with $\alpha_1 \ge \cdots \ge \alpha_r$, and let the idiosyncratic edge satisfy $\lambda_{r+1} \asymp_P N^{\alpha_e}$—with $\alpha_e = 0$ in the canonical scaling in which the idiosyncratic spectrum is bounded (Section 1). Then*

$$n = \lambda_r/\lambda_{r+1} \asymp_P N^{\alpha_r - \alpha_e},$$

*and Assumption **B4** ($n \gg T^{1/2}$) is equivalent to*

$$\alpha_r - \alpha_e > \frac{\ln T}{2\ln N}, \qquad \textit{i.e.} \qquad \alpha_r - \alpha_e > \tfrac{1}{2} \ \textit{ in balanced panels } (T \asymp N).$$

With a bounded idiosyncratic spectrum ($\alpha_e = 0$) this reads simply $\alpha_r > \frac{1}{2}$.

B4 is not a joint restriction on (T, N) but a lower bound on the strength of the weakest retained factor. It holds throughout the classical strong-factor region ($\alpha_r = 1$) and across most of the usable weak-factor range, failing only for very weak factors ($\alpha_r \le \frac{1}{2}$)—precisely the regime in which PC inference is fragile and in which one wants the test near its power boundary rather than delivering a clean Gaussian size. Equivalently, it is a stationarity boundary: a near-unit-root factor compresses n relative to $\sqrt{T}$ (Online Appendix) and so degrades the approximation even when the loadings are strong. The borderline simulations of the Online Appendix corroborate the threshold numerically: at $\alpha_r = 0.5$ the realized $n/\sqrt{T}$ sits near a constant (≈ 0.2 at $N = 200$), exactly the $\alpha_r = \frac{1}{2}$ boundary.

Let $\lambda_1 \ge \lambda_2 \ge \cdots \ge \lambda_T$ denote the ordered eigenvalues of XX'. Throughout, $W = A'A$ denotes the Gram matrix of the transformation A, and $n \equiv \lambda_1/\lambda_2$ is the spectral gap (one-factor version of λ_r/λ_{r+1}).

Define the scaled idiosyncratic part

$$\Delta \;=\; \frac{e\,M_\lambda e'}{N}, \qquad M_\lambda = I_N - \frac{\Lambda\Lambda'}{\Lambda'\Lambda},$$

and the normalized factor vector

$$f \;=\; \frac{\Sigma_F^{1/2}\xi}{\sqrt{T}}.$$

Because the component of f that is not independent of Δ has norm $O_p(n^{-1/2})$, it does not contribute at the order we require (see the Online Appendix).

The asymptotic variance depends on the following limits, which we define precisely and then show how to estimate:

$$\rho_k \;=\; \text{plim}_{N,T\to\infty}\frac{1}{T}\,\text{tr}\big(\Delta^k\big)\,, \quad k=1,2,4, \tag{3.3}$$

$$\tau_k \;=\; \text{plim}_{N,T\to\infty}\frac{1}{T}\,\text{tr}\big(\Sigma_F W^k\big)\,, \quad k=0,1,2,3, \tag{3.4}$$

$$\tau_{ab} \;=\; \text{plim}_{N,T\to\infty}\frac{1}{T}\,\text{tr}\big(\Sigma_F W^a \Sigma_F W^b\big)\,, \quad a,b\in\{0,1,2\}. \tag{3.5}$$

Define the normalized ratios

$$\gamma_k = \frac{\tau_k}{\tau_1^k}, \qquad \gamma_{ab} = \frac{\tau_{ab}}{\tau_1^{a+b}}.$$

Note that $\tau_0 = T^{-1}\,\text{tr}\,\Sigma_F = 1$ by the normalization convention, so $\gamma_0 = 1$ and $\tau_{00} = T^{-1}\,\text{tr}\,\Sigma_F^2$.

The moments ρ_k are consistently estimated from the non-factor eigenvalues, $\hat\rho_k = T^{-1}\sum_{j\ge 2}(\lambda_j/N)^k$, and the τ_k, τ_{ab} by plugging the PC estimate $\hat\Sigma_F$ and the Gram matrix W into the trace functionals (3.4)–(3.5); we require $\hat\tau_k - \tau_k = o_P(T^{-1/2})$ $(k=1,2)$ and $\hat\tau_{ab} - \tau_{ab} = o_P(1)$, which hold under standard short-memory conditions on F because these are normalized traces rather than individual entries of Σ_F.

The test statistic is $S = 1 - \text{tr}[P_{\hat F}P_{\tilde F}]$, where $P_{\hat F}$ and $P_{\tilde F}$ are the orthogonal projections onto the column spaces of $\hat F$ and $\tilde F = A^{-1}\hat F_A$, respectively. By Theorem 2.1, $n^2 S = \|P_0 D_A \Phi_1 M_0\|_F^2 + O_p(n^{-1})$ under the null of correct specification, where $D_A \equiv A'\Phi_{A0}^{+}A - \Phi_0^{+}$ is a deterministic contrast operator, $\Phi_1 = \lambda_2^{-1} e M_\Lambda e'$, and P_0, $M_0 = I - P_0$ are the projections onto and orthogonal to the true factor space.

Theorem 3.2 (Asymptotic distribution, one-factor case)**.** *Let $r=1$. Under Assumptions A1-A3, B1-B4,*

$$\sqrt{T}\left[\left(\frac{\lambda_1}{N}\right)^2 S \;-\; \omega_1\right] \;\xrightarrow{d}\; N(0,\,\Omega), \tag{3.6}$$

where

$$\omega_1 = \rho_2(\gamma_2 - 1), \tag{3.7}$$

$$\begin{aligned}\Omega = {} & 2\big(\gamma_{22} - 4\gamma_2\gamma_{12} + 4\gamma_2^2\gamma_{11} + \gamma_2^2(\gamma_{00} - 4\gamma_{01}) + 2\gamma_2\gamma_{02}\big)\,\rho_2^2 \\ & + 2\,(\gamma_2 - 1)^2\big(\rho_4 - \rho_2^2\big)\,. \end{aligned} \tag{3.8}$$

The quantities $\rho_2, \rho_4, \gamma_2, \gamma_{00}, \gamma_{01}, \gamma_{02}, \gamma_{11}, \gamma_{12}, \gamma_{22}$ are defined in (3.3)–(3.5) above and are estimated consistently by the plug-ins above.

The closed-form variance (3.8) relies on the asymptotic free-independence of the idiosyncratic matrix Δ and the Gram matrix $W = A'A$, which holds for every deterministic dip in the recommended library (in particular whenever the cross-sectional covariance Σ is not co-diagonalizable with W). Three robustness extensions, developed in the Online Appendix, delimit the result. (i) the feasible plug-in $\hat{\Omega}$—formed directly from $\hat{\Sigma}_F$, W, and the eigenvalue moments—is consistent and the studentized statistic pivotal *whether or not* free-independence holds, so structured dips are admissible; (ii) relaxing Gaussianity of the factors and idiosyncratic terms leaves Ω unchanged up to a fourth-cumulant term $\kappa_F\Psi$ that is negligible for the smooth dips used throughout; and (iii) under weak idiosyncratic serial dependence the limit law holds verbatim for random dips, while a temporal dip co-structured with persistent idiosyncratic terms is handled by the parametric-bootstrap standardization used in the simulations.

Proposition 3.3 (Consistency against detectable alternatives)**.** *Fix a dip A satisfying **A3** and a nominal level $\alpha \in (0,1)$, and let*

$$Z = \sqrt{T}\,\frac{(\lambda_1/N)^2 S - \hat{\omega}_1}{\sqrt{\hat{\Omega}}}$$

be the studentized statistic of Theorem 3.2, the test rejecting when $Z > z_{1-\alpha}$. Suppose the working specification is misspecified at rank $r = 1$ in the detectable *sense that, as $N, T \to \infty$,*

$$\left(\tfrac{\lambda_1}{N}\right)^2 S \xrightarrow{P} q_*, \qquad \hat{\omega}_1 \xrightarrow{P} \omega_*, \qquad \hat{\Omega} \xrightarrow{P} \Omega_* \in (0,\infty), \qquad q_* > \omega_*. \tag{3.9}$$

Then $\Pr(Z > z_{1-\alpha}) \to 1$, so the test is consistent. The same conclusion holds for the omnibus statistic Z_{blk} of Section 3.2 and for the library-maximal statistic $\max_{A\in\mathcal{L}} Z_{\text{blk}}(A)$ over any fixed finite library $\mathcal{L}$.

Condition (3.9) has a transparent reading. Under H_0 the annihilation $M_0 A^{-1} P_{Aj} = 0$ forces the first-order term of S to cancel, leaving $S = O_P(n^{-2})$ centred at ω_1; under H_1 the

r-th direction is not cleanly separated, the annihilation fails, the first-order term survives, and S is inflated beyond n^{-2}, so $q_* = \operatorname{plim}(\lambda_1/N)^2 S > \omega_*$. The Monte-Carlo power traces this mechanism: weakening a factor shifts the whole distribution of Z rightward rather than merely fattening its tail.

3.2 Extension to the multi-factor case

For $r > 1$ the statistic $S = r - \operatorname{tr}[P_{\hat{F}} P_{\tilde{F}}]$ measures the stability of the whole r-dimensional factor space. As Appendix E shows, the second-order perturbation behind Theorem 2.1 carries over with the rank-one eigenprojector replaced by the rank-r group projector, so only the block gap $n = \lambda_r/\lambda_{r+1}$ enters (not the within-block spacings), and the block statistic decomposes into a sum of r asymptotically independent one-factor statistics—valid even under clustered eigenvalues, where individual eigenvectors are inconsistent.

Notation and assumptions. Let $\bar{F} = (\bar{f}_1, \ldots, \bar{f}_r)$ be an orthonormal basis of the factor space ($\bar{F}'\bar{F} = TI_r$), $P_0 = \bar{F}\bar{F}'/T$, $M_0 = I_T - P_0$, and $P_{0a} = \bar{f}_a\bar{f}_a'/T$, $M_{0a} = I_T - P_{0a}$; as before $W = A'A$, $\Delta = eM_\Lambda e'/N$. We maintain A1–A3 and B2–B4, the gap read at the block $n = \lambda_r/\lambda_{r+1}$, and replace B1 by its multi-factor analogue:

B1′ Each factor carries its own temporal covariance, $\bar{f}_a = \Sigma_{F,a}^{1/2}\xi_a$ with $\xi_a \sim N(0, I_T)$ independent across a and independent of e, and $T^{-1}\operatorname{tr}\Sigma_{F,a} = 1$.

Under B1′ these coordinates agree with $\bar{F}$ up to an $O_p(T^{-1/2})$ rotation, the only channel through which directions interact. With $\Lambda_{\text{blk}} = \operatorname{diag}(\lambda_1, \ldots, \lambda_r)$, direction a enters at scale $(\lambda_a/T)^2$, so under the common normalization $(\lambda_r/N)^2$ its summand carries the (consistently estimable) weight $(\lambda_r/\lambda_a)^2$.

Theorem 3.4 (Block reduction)**.** *Under Assumptions A1–A3, with P_0 the rank-r group projector,*

$$n^2 S = \|P_0 D_A \Phi_1 M_0\|_F^2 + O_p(n^{-1}), \qquad \Phi_1 = \lambda_{r+1}^{-1} e M_\Lambda e',$$

and only the block gap $n = \lambda_r/\lambda_{r+1}$ enters: no within-block term $1/(\mu_i - \mu_j)$, $i, j \le r$, appears.

Because the ranges of the P_{0a} are mutually orthogonal, $\|P_0 Z\|_F^2 = \sum_a \|P_{0a} Z\|_F^2$; writing $\Phi_1 = (N/\lambda_{r+1})\Delta$ this gives the exact decomposition

$$n^2 S = \Big(\frac{N}{\lambda_{r+1}}\Big)^2 \sum_{a=1}^{r} S_a + \mathcal{E}_T, \qquad S_a = \|P_{0a} D_A \Delta M_{0a}\|_F^2, \tag{3.10}$$

where the remainder $\mathcal{E}_T = \sum_a \big(\|P_{0a} D_A \Delta M_0\|_F^2 - \|P_{0a} D_A \Delta M_{0a}\|_F^2\big)$ collects the cross-direction terms.

Lemma 3.5 (Per-factor reduction, remainder, independence)**.** *Under Assumptions A1–A3, B1′–B4 and $n \gg T^{1/2}$: (i) each summand S_a reduces to the one-factor statistic of Section 3.1 for direction a (Σ_F replaced by $\Sigma_{F,a}$) carried at the common normalization, so that S_a centres at $(\lambda_r/\lambda_a)^2\omega_1^{(a)}$, up to a term that is $o_p(1)$ after the $\sqrt{T}$ standardization below; (ii) $\sqrt{T}\,(\lambda_r/N)^2\mathcal{E}_T = o_p(1)$; (iii) the standardized summands are asymptotically independent across a.*

We distinguish two regimes according to the factor temporal covariances. Write $\omega_1^{(a)}$ and $\Omega^{(a)}$ for the one-factor centring (3.7) and variance (3.8) evaluated at $\Sigma_{F,a}$ (only the γ's carry the factor index; ρ_2, ρ_4 are common).

Theorem 3.6 (Homogeneous regime, Case H)**.** *Suppose $\Sigma_{F,a} = \Sigma_F$ for all a and write ω_1, Ω for the common one-factor centring and variance. Under Assumptions A1–A3, B1′–B4 and $n \gg T^{1/2}$, with $\Lambda_{\mathrm{blk}} = \mathrm{diag}(\lambda_1, \dots, \lambda_r)$,*

$$\sqrt{T}\Big[\Big(\tfrac{\lambda_r}{N}\Big)^2 S - \omega_1 \sum_{a=1}^{r}\Big(\tfrac{\lambda_r}{\lambda_a}\Big)^2\Big] \xrightarrow{d} N\Big(0,\ \Omega\sum_{a=1}^{r}\Big(\tfrac{\lambda_r}{\lambda_a}\Big)^4\Big),$$
$$Z_{\mathrm{blk}} = \frac{\sqrt{T}\big[(\lambda_r/N)^2 S - \hat{\omega}_1 \sum_a (\lambda_r/\lambda_a)^2\big]}{\sqrt{\hat{\Omega}\sum_a (\lambda_r/\lambda_a)^4}} \xrightarrow{d} N(0,1),$$

with $\hat{\omega}_1, \hat{\Omega}$ the plug-ins from the common Σ_F (the block-estimable temporal covariance) and λ_a the observed eigenvalues. When the factor strengths are equal, $\lambda_1 = \cdots = \lambda_r$, every weight is unity and the statement reduces to $\sqrt{T}[(\lambda_r/N)^2 S - r\omega_1] \xrightarrow{d} N(0, r\Omega)$.

Theorem 3.7 (Heterogeneous regime, Case K)**.** *Suppose the $\Sigma_{F,a}$ are factor-specific and the relevant trace functionals are estimable at rate $o_p(T^{-1/2})$. Under Assumptions A1–A3, B1′–B4 and $n \gg T^{1/2}$, with $\Lambda_{\mathrm{blk}} = \mathrm{diag}(\lambda_1, \dots, \lambda_r)$,*

$$\sqrt{T}\Big[\Big(\tfrac{\lambda_r}{N}\Big)^2 S - \sum_{a=1}^{r}\Big(\tfrac{\lambda_r}{\lambda_a}\Big)^2 \omega_1^{(a)}\Big] \xrightarrow{d} N\Big(0,\ \sum_{a=1}^{r}\Big(\tfrac{\lambda_r}{\lambda_a}\Big)^4 \Omega^{(a)}\Big),$$
$$Z_{\mathrm{blk}} = \frac{\sqrt{T}\big[(\lambda_r/N)^2 S - \sum_a (\lambda_r/\lambda_a)^2 \hat{\omega}_1^{(a)}\big]}{\sqrt{\sum_a (\lambda_r/\lambda_a)^4 \hat{\Omega}^{(a)}}} \xrightarrow{d} N(0,1).$$

Equivalently, each direction is normalized by its own strength: $U_a := \sqrt{T}\big[(\lambda_a/N)^2 S_a - \omega_1^{(a)}\big] \xrightarrow{d} N(0, \Omega^{(a)})$ are asymptotically independent (Lemma 3.5(iii)), so the joint limit is $(U_1, \dots, U_r) \xrightarrow{d} N(0, \mathrm{diag}(\Omega^{(1)}, \dots, \Omega^{(r)}))$ and the block limits above are the Λ_{blk}-weighted contraction of this diagonal Gaussian. Theorem 3.6 is the special case $\Sigma_{F,a} \equiv \Sigma_F$.

Remark 3.8 (The eigenvalue weights are not optional)**.** The weights $(\lambda_r/\lambda_a)^2$ (centring) and $(\lambda_r/\lambda_a)^4$ (variance) collapse to unity only when the block eigenvalues are equal; they are not implied by $n \gg \sqrt{T}$, which restricts the block-to-noise gap but not the within-block ratios. Being consistently estimated even when the eigenvectors are not, they keep the statistic feasible in both regimes and improve finite-sample size when the strengths are spread.

Case H is stated separately because its variance needs only the block-estimable Σ_F, remaining feasible even when the individual directions are not; the two regimes are genuinely different, as the following boundary makes precise.

Proposition 3.9 (Identification boundary)**.** *If the factors are spectrally clustered (some within-block gaps are $O_p(1)$, so individual eigenvectors are inconsistent) and the $\Sigma_{F,a}$ are heterogeneous and not individually estimable, then $\sum_a \Omega^{(a)}$ is not identified: it depends on $\sum_a \mathrm{tr}(\Sigma_{F,a} W^a \Sigma_{F,a} W^b)$, which is not a function of the block sum $\sum_a \Sigma_{F,a}$. In this regime only Case H is feasible; in the well-separated regime the $\Sigma_{F,a}$ are estimable and Case K applies.*

3.3 A log-transformed statistic

A variance-stabilizing log transform of the level statistic $Q \equiv (\lambda_r/N)^2 S$, $Z^{\log}_{\mathrm{blk}} = \sqrt{T}\,\hat{\bar{\omega}}\,(\log Q - \log \hat{\bar{\omega}})/\sqrt{\hat{\bar{\Omega}}} \xrightarrow{d} N(0,1)$ (with $\bar{\omega}, \bar{\Omega}$ the centring and variance of Theorem 3.7), improves the small-sample normal approximation while leaving the first-order theory and power unchanged. We retain the level statistic as the default and develop the log alternative, which is better sized but slightly less size-adjusted-powerful (See Online Appendix).

3.4 The number of stable factors

The omnibus test of Section 3.2 is run for a given working number of factors r: it asks whether the r-dimensional PC factor space is stable under the transformation library. Applying it sequentially along $r = 1, 2, \ldots$ therefore yields a natural by-product, the number of stable factors

$$\hat{r}_{\mathrm{dip}} = \max\{\, r : \text{ the } r\text{-factor specification is not rejected} \,\},$$

the largest number of factors whose PC space is robust to the admissible data representations. Because successive factors are progressively weaker, and hence more fragile, $\hat{r}_{\mathrm{dip}}$ isolates the factors that are robustly identified rather than merely present.

The sequence of tests also delivers a set with a coverage interpretation. Let $r_s = \#\{j : \alpha_j > 1/2\}$ be the population number of stable factors—those whose eigen-direction is separated enough ($n_j \gg \sqrt{T}$, Lemma 3.1) to admit the Gaussian null law of Theorem 3.2.

Because the omnibus test controls asymptotic size under the stable-factor null, the non-rejected set $\widehat{\mathcal{R}} = \{r : \text{the } r\text{-factor specification is not rejected at level } \alpha\}$ is an asymptotic level-$(1-\alpha)$ confidence set for r_s: $\Pr(r_s \in \widehat{\mathcal{R}}) \to 1-\alpha$ whenever the ladder of tests is read at a common level. Reporting $\widehat{\mathcal{R}}$—rather than only the point summary $\hat{r}_{\text{dip}} = \max \widehat{\mathcal{R}}$—conveys the sampling uncertainty in the stable count directly, and sidesteps the need for a separate consistency argument. We use this confidence-set reading throughout.

These estimators answer a different question. The information criteria of Bai and Ng (2002), the eigenvalue-ratio estimators of Ahn and Horenstein (2013), and the edge-distribution methods of Onatski (2009, 2010) estimate *how many* factors are present; $\hat{r}_{\text{dip}}$ measures for how many the PC inference is *robust*. They coincide under strong, well-separated factors and diverge under weak ones ($\hat{r}_{\text{dip}} \leq \hat{r}_{\text{IC}}$), so we recommend reporting $\hat{r}_{\text{dip}}$ alongside the usual estimates.

A Monte Carlo comparison of $\hat{r}_{\text{dip}}$ against these selectors is reported in the Online Appendix.

4 Simulations

We assess the test and the uncertainty-assessment procedures by Monte Carlo; the designs and full results are in the Online Appendix. In brief, the feasible simulation-standardized omnibus test is correctly sized (0.036–0.063 at nominal 5%) across i.i.d., ARMA, and heterogeneous factor designs and every sample size; power against weak-factor misspecification rises steeply, from about 0.12 under correct specification to near one as factors weaken, and grows with N; reliability is governed by the realized gap $n/\sqrt{T}$ and degrades only for near-unit-root factors; and three further experiments confirm robustness to non-Gaussianity, a near-differencing diagnostic for the Onatski and Wang (2021) spurious-factor problem, and a clean separation of the stable-factor count from the variance-based number of factors.

5 Assessing PC Uncertainty

Beyond testing the standard specification, factor dipping provides a direct, model-free way to visualize the PC uncertainty that weak factors induce and that standard Bai (2003) intervals tend to understate. The idea is simple: a transformation A that leaves the factor space intact under correct specification but perturbs it under weak factors turns the family of admissible A's into a family of equally defensible factor estimates, and the spread of these estimates is itself a measure of uncertainty—a representation-sensitivity band rather than a confidence interval (Section 6). We assess it at two levels.

Factor estimates. Apply dipping across a single transformation of interest or a library $\mathcal{L}$ (Section 6) to obtain $\{\tilde{F}_\ell\}$ and compare with $\hat{F}$: under the standard specification they coincide with $\hat{F}$ up to $O_p(n^{-2})$ and lie within the Bai (2003) band, while weak factors make them spread.

Downstream quantities. Factor estimates are usually intermediate inputs to FAVAR impulse responses, forecasts, or risk premia, in which they enter more than once. We therefore propagate the dips: for each $\tilde{F}_\ell$ we substitute it for $\hat{F}$ from the first step in which factors appear and recompute the whole pipeline, yielding $\{\tilde{R}_\ell\}$ to compare against $\hat{R}$.

We summarize the uncertainty two ways: the envelope, the pointwise range of the dipped quantities across the library (a conservative band); and a visualization overlaying the original estimate, its Bai (2003) band, and all dipped quantities, so the reader can judge whether the dipping-induced spread exceeds the nominal uncertainty. Procedures 1 and 2 summarize the two levels.

The downstream assessment is related to the generated-regressor problem: uncertainty in the factors propagates to any quantity built on them, with an impact that varies across applications, and we recommend assessing it before interpreting PC-based results.

Procedure 1 Uncertainty assessment for the factor estimates

1: Compute the PC factor estimate $\hat{F}$ and its standard Bai (2003) confidence band.
2: Choose a single A of interest or a library $\{A_\ell\}$ (Section 6) and apply factor dipping to obtain the dipped estimates $\{\tilde{F}_\ell\}$.
3: Report the envelope $[\min_\ell \tilde{F}_\ell,\ \max_\ell \tilde{F}_\ell]$ and overlay $\hat{F}$, its band, and all $\{\tilde{F}_\ell\}$ in a single figure.

Procedure 2 Uncertainty assessment propagated to downstream results

1: Compute $\hat{F}$ and the downstream result $\hat{R}$ (e.g. a FAVAR impulse response, forecast, or risk premium) together with its standard uncertainty measure.
2: Choose a library $\{A_\ell\}$ and apply factor dipping to obtain $\{\tilde{F}_\ell\}$.
3: For each ℓ, replace $\hat{F}$ by $\tilde{F}_\ell$ at the first step in which factors enter and recompute the whole pipeline to obtain $\tilde{R}_\ell$.
4: Report the envelope $[\min_\ell \tilde{R}_\ell,\ \max_\ell \tilde{R}_\ell]$ and overlay $\hat{R}$, its uncertainty measure, and all $\{\tilde{R}_\ell\}$ in a single figure.

The envelope does not cry wolf. For the envelope to be useful it must be quiet when the factors are strong—manufacturing no spurious uncertainty under correct specification—and widen only where the representation genuinely matters. We verify both. In a three-factor design ($T = N = 200$, 250 replications) we hold the first two factors strong ($\alpha = 1$) and vary the strength α_3 of the third, applying a representative subset of the library of

Section 6 (autoregressive $A(\rho)$ with $\rho = \pm 0.3, \pm 0.5$; KMS $a = \pm 0.3, \pm 0.5$; three random $\tilde{A}$ with $\sigma_t \in [1, 2]$). Table 3 reports the minimum canonical correlation between the dipped and PC factor spaces and the normalized pointwise envelope width (unit-norm factor estimates, scaled by $\sqrt{T}$) for a strong factor and for the weakened one.

Table 3: No-false-alarm check for the recommended-library envelope ($r = 3$, factors 1–2 strong, factor 3 at strength α_3; $T = N = 200$, 250 replications).

α_3 (factor 3)	min. canon. corr. (space)	envelope width, strong factor 1	envelope width, weak factor 3
1.0 (all strong)	1.000	0.107	0.113
0.8	1.000	0.105	0.121
0.6	0.999	0.104	0.194
0.4	0.986	0.095	0.501

Under all-strong factors the envelope is tight—the dipped and PC spaces are numerically identical (minimum canonical correlation 1.00) and widths are small and equal across factors—so dipping invents no uncertainty where none is warranted. As the third factor weakens, its envelope widens monotonically (roughly fourfold from $\alpha_3 = 1$ to 0.4) while the strong factor's is unmoved. The band thus stays quiet under correct specification and, when it widens, points to the specific factor whose PC estimate is representation-sensitive.

6 A Library of Dips

The only requirement on a dip is Assumption **A3**: $\|A\| = O_P(1)$ and $\|A^{-1}\| = O_P(1)$. A dip may be chosen in two paths—as a specific representation whose influence one wants to probe, or as a member of a recommended library spanning the main families of data representation, whose envelope measures the sensitivity of PC inference across the representations an applied researcher might defensibly adopt.

A transformation of substantive interest. A specific pre-PCA operation can itself be the question, but only operations satisfying Assumption **A3** are admissible—for example an invertible smoother or a band-pass-type filter with parameters inside the unit disk (Hodrick and Prescott, 1997; Stock and Watson, 2016)—applied as AX on the temporal side, or in cross-sectional analogue as XB with B bounded and boundedly invertible. Dipping with such an A then tests precisely whether that representation choice matters.

A recommended library. When the goal is to gauge the overall PC uncertainty that data-representation choices can induce, we recommend a library that spans the operations an applied researcher actually performs upstream of PCA. Each family below is bounded with bounded inverse, so satisfies **A3** by construction, and each corresponds to a recognizable

pre-processing choice, so that the resulting envelope has a concrete interpretation as the spread of PC inference across those choices.[4]

1. Autoregressive / quasi-differencing filters: $A(\rho) = I - \rho L$, $|\rho| < 1$ ($\|A(\rho)^{-1}\| \to (1 - |\rho|)^{-1}$): pre-whitening and detrending; as $\rho \to 1$ this approaches inadmissible first-differencing, the near-differencing probe of the Online Appendix.

2. Exponential-smoothing (Kac–Murdock–Szegő): Toeplitz $A_{ij} = a^{|i-j|}$, $a \in (-1, 1)$, bounded uniformly in T: the family used throughout our illustrations.

3. Moving-average smoothers and band-pass filters: (Hodrick–Prescott, Baxter–King, Christiano–Fitzgerald) in bounded, boundedly invertible form (Hodrick and Prescott, 1997; Stock and Watson, 2016).

4. Averaging and temporal aggregation: row-stochastic local-averaging operators blending a window-m moving average with the identity, bounded and invertible when the blend keeps the smallest eigenvalue away from zero.

5. Randomly generated bounded matrices: $\tilde{A} = U_1 D_\sigma U_2'$ with U_1, U_2 random orthogonal and σ_t drawn on a short interval such as $[1, 2]$: agnostic stability probes, free of any temporal structure.

For routine use we recommend a large, fixed library that samples all five families over mild parameter ranges, so that the envelope reflects the range of admissible representations rather than a single tuning. [5] The library is fixed in advance and not optimized against the data, which is what makes the resulting envelope a genuine robustness object rather than a data-snooped one. Because it is impractical to tabulate 65 columns, the empirical displays below report the envelope (the range of the dipped quantity over the library) together with a representative handful of individual members.

The envelope is a representation-sensitivity band, not a confidence interval: at each point it reports the range of equally defensible dipped estimates as the representation varies over the fixed library. Overlaid on the Bai (2003) sampling band the comparison is diagnostic, not

[4] We list the temporal-side operators AX, and each can have a cross-sectional analogue XB acting on the loadings. Cross-sectional standardization and weighting enter as XB with B bounded and invertible—admissible precisely when the scale factors are bounded away from zero and infinity (Greenaway-McGrevy et al., 2012; Boivin and Ng, 2006).

[5] Our default combines exponential-smoothing (KMS) dips $a = \pm 0.1, \ldots, \pm 0.6$ (12 members); autoregressive / quasi-differencing dips $A(\rho)$ with $\rho = \pm 0.1, \ldots, \pm 0.7$ (14); moving-average smoothers $I + w(L + L')$ with $w = \pm 0.1, \ldots, \pm 0.4$ (8); row-stochastic local averages over windows $m \in \{1, 2\}$ with $c \leq 0.5$ (7); and a batch of random $\tilde{A} = U_1 D_\sigma U_2'$ with $\sigma_t \in [0.5, 2]$ (24)—65 dips in all. Every member is mild: $\|A\|$ and $\|A^{-1}\|$ are bounded, with condition number no larger than that of KMS(0.6) (≈ 16), so the comparison across the library is like-for-like in perturbation magnitude.

like-for-like—a dipped spread exceeding the nominal band signals that the point estimate is fragile to representation choices the sampling band ignores. Because the library is fixed and mild, a tight envelope certifies robustness and a wide one localizes which factors and downstream objects are representation-sensitive.

7 Empirical Illustrations

We now demonstrate factor dipping on three PC-based analyses: the quantile factor model of Chen et al. (2021), the three-pass risk-premium estimator of Giglio and Xiu (2021), and a factor-augmented VAR (Bernanke et al., 2005). Throughout, dipping is applied across the full recommended library (65 dips), and we report the envelope with a representative handful of members. [6]

7.1 Quantile factor model

Our first illustration revisits the quantile factor model (QFM) of Chen et al. (2021), estimated on the FRED-MD macroeconomic panel, in which factors are extracted at each quantile so that the tail factors are the weakest and the misspecification-induced uncertainty is largest. We apply Procedure 1 across the full recommended library to their eight estimated PC factors. Figure 2 overlays the dipped estimates on the original PC factor and its standard Bai (2003) 90% band. The pattern is exactly the one the invariance theorem predicts: the dipped estimates are visually indistinguishable from the original for the first few factors—whose eigenvalue gaps are wide—and fan out progressively for the later ones, with the seventh and eighth factors' dipped estimates departing markedly and spending much of the sub-period *outside* the nominal 90% band. The standard interval, computed at a single representation, thus understates the true uncertainty in the weak factors by a wide margin, precisely where quantile factor analysis is most often used to read tail dynamics. A companion regression makes the point numerically (Online Appendix). Chen et al. (2021) report that the QFA factors are closely spanned by the eight PC factors near the median and upper quantiles, with R^2 near unity, and read this as evidence that PCA recovers the QFA structure there.

[6] Before applying the diagnostic we verify, on all three panels, that the library does what Theorem 2.1 requires: it leaves the strong, well-separated factors intact and perturbs only the weak ones. The Online Appendix reports, for each PC factor, the minimum canonical correlation between the original and dipped estimate over all 65 dips. The pattern is monotone in the eigenvalue gap $n_j = \lambda_j/\lambda_{j+1}$: every well-separated factor ($n_j > 1.5$) is preserved to canonical correlation between 0.96 and 1.00 by every member of the library, while only the near-degenerate factors ($n_j \approx 1$) move. No dip in the library spoils a strong factor; the instability the diagnostic reports below is confined to the weak factors it is designed to flag, exactly the $O_p(n^{-2})$ behaviour of Theorem 2.1.

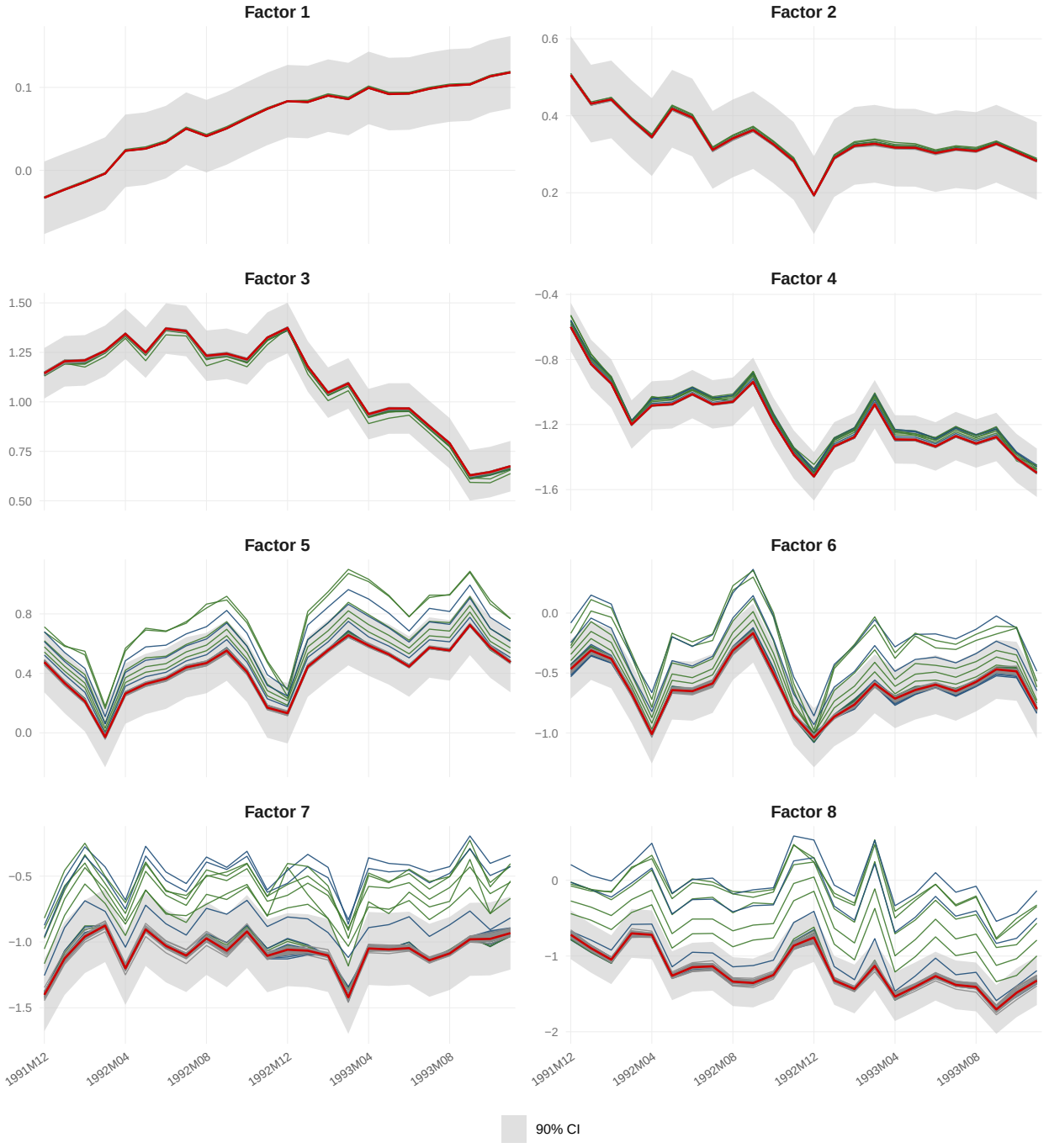


Figure 2: The eight QFM PC factor estimates (red) with 90% confidence intervals (grey band) and the dipped estimates across the full recommended library (65 dips), coloured by family (blue for positive-sign structured dips, green for negative-sign structured, grey for averaging and random); FRED-MD monthly data, a short sub-period shown for clarity. The dipped estimates stay close to the original for the strongest factors and increasingly exceed the standard 90% band for the weaker ones.

Regressing the same QFA factors instead on two dipped PC spaces ($\mathbf{A}_T(\pm 0.6)$) shows the alignment survives only for the leading factor: at $\tau = 0.50$ the fits of the third-to-fifth QFA factors fall from 0.96, 0.95, 0.92 under the original PC estimates to 0.45, 0.23, 0.25 under $\mathbf{A}_T(-0.6)$ (and the second factor from 0.98 to 0.54 under $\mathbf{A}_T(0.6)$), with the same collapse at $\tau = 0.75$, while the leading factor stays above 0.89 throughout. The close PCA–QFA correspondence at these quantiles is therefore an artefact of one representation—exactly the fragility the overlay in Figure 2 displays.

7.2 Three-Pass Risk Premium Estimator

We applied factor dipping to the three-pass principal-component factor model of Giglio and Xiu (2021) as implemented for their Table B1.[7] The three-pass estimator recovers the premium $\hat{\gamma}_j$ of a candidate factor g_j by extracting $p = 7$ PC factors $\hat{F}$ from test-asset returns, running Fama–MacBeth regressions to recover PC premia, and projecting g_j onto the PC space. For each KMS dip $\mathbf{A}_T(a)$, $a \in \{\pm 0.3, \pm 0.5, \pm 0.6\}$, we obtain $\tilde{F}$ and re-run the whole estimator with $\tilde{F}$ in place of $\hat{F}$. The resulting premia across all seven specifications are reported in Table 4.[8]

The table separates robust from fragile premia, and the separation is graded precisely by the spanning strength R^2_G—the pass-3 R^2 of the candidate factor on the PC space. The strongly priced equity and intermediary factors move only modestly under dipping, and the band width tracks R^2_G monotonically: the market factor ($R^2_G = 99.9\%$) stays within $[0.45\%, 0.53\%]$ of its original 0.51% across the entire 65-dip library; as the spanning strength falls the band widens smoothly—HML (82.9%) to $[-0.13\%, 0.38\%]$, momentum (89.9%) to $[-0.16\%, 0.79\%]$, the He intermediary factor (69.9%) to $[0.07\%, 0.97\%]$—yet every one of these movements is economically immaterial and statistically indistinguishable from the original, two orders of magnitude smaller than the placebo swings below. Factor dipping thus does not manufacture fragility where the factor space is well identified.

By contrast, the weakly exposed factors ($R^2_G \lesssim 6\%$) are genuinely fragile: their premia swing by economically enormous amounts and reverse sign across the library—New York temperature from -51% to $+30\%$, Sunspots from -65% to $+90\%$—while their t-statistics stay far below conventional thresholds, a symptom of the near-singular Pass-3 mapping when a candidate loads negligibly on the latent space. Dipping provides a credibility check, quiet where the candidate factor is well identified by the PC space and loud where it is not.

[7] The replication uses 407 monthly observations and 647 test assets (excess returns, dropping any series with missing data), with $p = 7$ latent principal-component (PC) factors and Newey–West standard errors at lag truncation $q = 3$. These choices match the baseline specification of Giglio and Xiu (2021).

[8] We orthonormalize $\tilde{F}$ ($\tilde{F}'\tilde{F}/T = I_r$) before re-running the estimator, ensuring that any change reflects a change of factor space.

Table 4: Risk-premium estimates $\hat{\gamma}$ (%) under factor dipping (replication of Giglio and Xiu 2021, Table B1). "Original" refers to the baseline three-pass estimator with $p = 7$ latent PC factors and Newey–West standard errors ($q = 3$ lags). The six dipping columns replace the SVD-based PC factors with dipped factors constructed via the KMS matrix $A(a)$, $A_{ij} = a^{|i-j|}$, for $a \in \{+0.3, -0.3, +0.5, -0.5, +0.6, -0.6\}$. Orange cells flag deviations exceeding $\max(0.15\,\text{pp},\ 50\%\,|\hat{\gamma}_{\text{orig}}|)$. Red cells flag a sign change relative to the original. R^2_G denotes the pass-3 R^2 of the candidate factor on the PC space.

	Original ($p = 7$)				**Dipped $\hat{\gamma}$ (%) by KMS parameter a**					
Factor	$\hat{\gamma}$	S.E.	t	R^2_G	$+0.3$	-0.3	$+0.5$	-0.5	$+0.6$	-0.6
Standard Equity Factors										
Market	0.508	(4.687)	0.108	99.9%	0.514	0.477	0.519	0.496	0.510	0.517
SMB	0.198	(3.186)	0.062	97.2%	0.114	0.296	0.084	0.334	0.106	0.232
HML	0.202	(3.082)	0.065	82.9%	0.172	0.273	0.169	0.217	0.306	0.250
MOM	0.486	(4.587)	0.106	89.9%	0.495	0.475	0.480	0.424	0.538	0.471
Factor Zoo										
RMW	0.217	(2.159)	0.101	71.0%	0.030	0.182	0.056	0.115	0.096	0.113
CMA	0.139	(1.695)	0.082	58.9%	0.154	0.163	0.101	0.218	0.200	0.254
BAB	0.570	(2.825)	0.202	47.6%	0.334	0.720	0.216	0.533	0.161	0.504
QMJ	0.055	(2.591)	0.021	84.0%	**−0.033**	0.067	**−0.076**	0.043	**−0.049**	0.042
Liquidity	0.367	(2.129)	0.172	12.0%	0.234	0.388	0.251	0.158	0.183	0.178
Financial Intermediaries										
Interm. (He)	0.605	(6.090)	0.099	69.9%	0.110	0.853	0.261	0.713	0.407	0.630
Interm. (Adrian)	0.722	(3.177)	0.227	51.4%	0.790	0.896	0.636	0.867	0.901	0.870
Weather / Non-Priced										
NY temp.	−0.692	(89.63)	−0.008	0.8%	−9.149	−5.977	−24.628	−20.803	−15.503	−18.682
Global temp.	0.047	(2.419)	0.019	2.1%	**−0.206**	**−0.049**	**−0.048**	0.082	0.314	0.151
El Niño	0.414	(6.592)	0.063	1.6%	0.422	1.058	0.031	0.272	1.152	0.504
Sunspots	4.013	(252.4)	0.016	0.9%	**−27.798**	**−15.848**	**−5.237**	**−13.754**	36.223	8.371
Macro / IP Growth										
IP growth	−0.009	(0.048)	−0.184	2.2%	−0.010	−0.010	−0.010	−0.014	−0.010	−0.012
JLN Macro Uncertainty										
Macro PC1	3.260	(16.94)	0.192	2.4%	0.307	2.543	1.967	3.756	3.285	4.039
Macro PC2	−0.879	(14.93)	−0.059	4.3%	−0.943	−2.738	−1.030	−1.970	−1.140	−1.341
Macro PC3	−1.254	(18.06)	−0.069	6.0%	−4.478	**0.708**	−4.230	**0.929**	−2.189	−0.117
Consumption										
Cons. growth	0.004	(0.077)	0.055	3.8%	**−0.009**	0.010	**−0.011**	0.012	**−0.009**	0.009
Stockholder cons.	0.171	(0.503)	0.340	2.5%	0.064	0.250	0.128	0.142	0.117	0.228

7.3 Factor-Augmented Vector Autoregression (FAVAR)

Our last illustration is the impulse responses of a factor-augmented VAR (Bernanke et al., 2005) to a monetary-policy shock, with factors extracted by PC from a large monthly macroeconomic panel and entered, with the funds rate, into a VAR. We take the Bernanke et al. (2005) specification as given. These are near-integrated levels, which place the omnibus test in the over-rejection regime of the Online Appendix, so we rely on the representation-sensitivity envelope rather than the reject/accept test. We do not use dipping to choose the number of factors: Bernanke et al. (2005) deliberately leave k open, reporting three- and five-factor models and noting that standard criteria (Bai and Ng, 2002) do not settle it. Our question is complementary—taking their factors as given, what does dipping add? Including more, weaker factors explains more of the panel but strains the specification behind the fixed-k bootstrap band, injecting a representation-induced uncertainty the band misses; dipping reveals that extra source, response by response, rather than counting factors.

We reproduce the responses for $k = 1, \ldots, 8$ factors and apply factor dipping (Procedure 2) across the full recommended library of Section 6, substituting each dipped factor estimate for the original at the first step and recomputing the entire two-step pipeline. The first thing dipping establishes is that the exercise is not about the number of factors at all: the PC factor space is nearly invariant to the representation. The minimum canonical correlation between the original and dipped factor spaces over all 65 dips is 1.00 for the first three factors, 0.999 and 0.991 for the fourth and fifth, and drops below 0.9 only at the seventh and eighth. Dipping thus does not quarrel with the Bernanke et al. (2005) factor count—the space those factors span is robust. What dipping adds lives entirely downstream, in how the structural impulse responses built on that space depend on which admissible representation is used.

Dipping grades the impulse responses by representation-robustness. Figure 3 shows three responses spanning the range. The *exchange rate* is representation-robust—its dipped responses stay inside the Bernanke et al. (2005) band at every k, so dipping certifies it. The *S&P 500* is representation-fragile: already at $k = 3$ the dips disagree on the sign of the longer-horizon equity response, and at $k = 5$ neither its sign nor magnitude is pinned down. The most consequential case is the *CPI*: the "price puzzle"—a counter-intuitive rise of prices after a contractionary shock—motivated Bernanke et al. (2005), who argued that factor augmentation mitigates it; dipping shows that on this panel whether the puzzle appears is itself a matter of representation, since the dips disagree most over the first year (where the puzzle lives) and at $k = 5$ smear a pronounced positive price hump across a wide range. Even the policy instrument is not immune—the funds rate's own long-horizon response is tightly pinned at $k = 1$ but ambiguous by $k = 3$. Response by response, dipping separates the

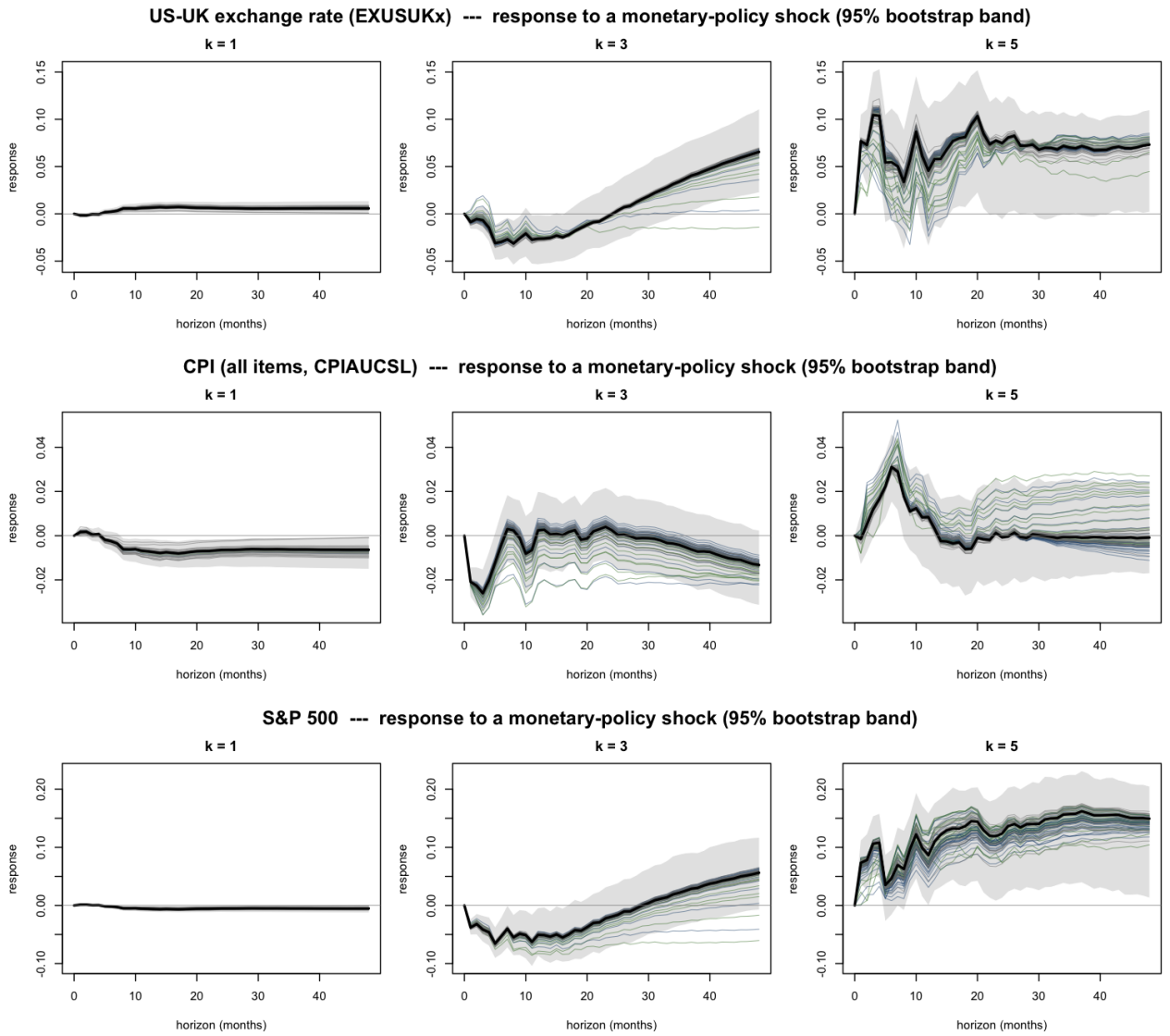


Figure 3: Dipped impulse responses to a monetary-policy shock from FAVARs with $k = 1, 3, 5$ factors (left to right): the US–UK exchange rate (top), the CPI (middle), and the S&P 500 (bottom). Each panel shows the original estimate (black), its 95% Bernanke et al. (2005) bootstrap band (grey), and the dipped estimates across the recommended library (blue/green for positive/negative structured dips, grey for random). The exchange rate is representation-robust, the S&P 500 fragile, and the CPI interpretation-critical—whether the "price puzzle" appears depends on the representation.

conclusions one may draw robustly from those that hold only under a single representation.

The sensitivity is systematic and grows with k. Across all 116 responses (Online Appendix) robustness tracks how heavily a response leans on the weaker, later factors: exchange rates, deflators, and the short rate are robust; the long end of the term structure and industrial production are shape-robust but magnitude-sensitive; and money/credit aggregates, equities (whose fan straddles zero), the labour market, and headline prices are fragile, their fans widening from $k = 3$ to $k = 5$. The mechanism is not the factor space moving—its span is preserved to canonical correlation ≥ 0.99 through five factors—but that each added factor is weaker, so its dipped direction wobbles more and the near-integrated VAR amplifies the wobble; a caveat sharpens the reading, since a recursive (Cholesky) identification is not invariant to a within-span rotation, so part of the fan reflects the identification scheme's sensitivity to the dipped basis rather than PC-estimation uncertainty alone. Both are representation effects a fixed-specification band ignores, widest for the fragile responses, and give a concrete reason to prefer the parsimony Bernanke et al. (2005) already favoured. Dipping thus adds a per-response credibility check, not a count of factors.

8 Conclusion

Factor dipping is a simple procedure that generates alternative factor estimates. Under the standard specification, the dipped estimates span essentially the same space as the PC estimates, with deviations of order $O_p(n^{-2})$. Significant departures from this near-invariance are evidence against the specification validity, and a warning that conventional uncertainty measure can no longer be relied upon. Because eigenvalue spectra in macroeconomic and financial panels routinely lack a sharp gap, this matters in practice: our applications show that standard intervals can badly understate the uncertainty weak factors inject, and that data pre-processing—itself a linear transformation—can implicitly destabilize PC inference unnoticed.

Factor dipping needs no additional data and negligible computation, yet is effective. We recommend applying it across a library of transformations before drawing inference from any PC-based analysis, reporting the envelope alongside point estimates much as robust standard errors accompany a regression today. It extends naturally to matrix and tensor factor models (Wang et al., 2019; Chen et al., 2022), to PCA of the estimated quadratic covariation matrix (Pelger, 2019), and to Risk-Premium PCA (RP-PCA) (Lettau and Pelger, 2020).

A Proof of Theorem 2.1

This appendix carries out the perturbation analysis summarized in Section 2, using the notation introduced there: $\Phi = \Phi_0 + \Phi_1/n$, the projections P_0, M_0, P_i, eigenvalues μ_i, the orthonormal basis $\bar{\mathcal{F}}$, the Moore–Penrose inverse $\Phi_0^+ = \sum_{i \le r} \mu_i^{-1} P_i$, and the transformed quantities carrying a subscript A.

A.1 First-order expansion and the invariance

The first-order formulae (2.3) and (2.14) of Kato (1980) give

$$P_{\hat{F}} = P_0 + \frac{1}{n} \sum_{i=1}^{r} P_i^{(1)} + \delta_1, \qquad \delta_1 = O_P(n^{-2}), \tag{A.1}$$

where $P_i^{(1)} = -P_i \Phi_1 S_i - S_i \Phi_1 P_i$ and S_i is the reduced resolvent of Φ_0 at μ_i,

$$S_i = \sum_{j=1,\, j \neq i}^{r} \frac{1}{\mu_j - \mu_i} P_j - \frac{1}{\mu_i} M_0.$$

Assumption **A1** keeps the r-th eigenvalue of Φ away from zero while the $(r+1)$-th vanishes, whence $\left\| \sum_{i=1}^{r} P_i^{(1)} \right\| = O_P(1)$ (via the contour-integral representation (2.8), p. 76 of Kato, 1980). Writing $\Phi_A = A \Phi A'$, $\Phi_{Ai} = A \Phi_i A'$ and $\Phi_A = \Phi_{A0} + \Phi_{A1}/n$, the transformed projection admits the same expansion

$$P_{\hat{F}_A} = P_{A0} + \frac{1}{n} \sum_{i=1}^{r} P_{Ai}^{(1)} + \delta_{A1}, \qquad \delta_{A1} = O_P(n^{-2}), \tag{A.2}$$

with

$$P_{Ai}^{(1)} = -P_{Ai} \Phi_{A1} S_{Ai} - S_{Ai} \Phi_{A1} P_{Ai}, \qquad S_{Ai} = \sum_{j \neq i} \frac{1}{\mu_{Aj} - \mu_{Ai}} P_{Aj} - \frac{1}{\mu_{Ai}} M_{A0}, \tag{A.3}$$

and $\left\| \sum_i P_{Ai}^{(1)} \right\| = O_P(1)$. Multiplying (A.2) by $A\bar{\mathcal{F}}$ on the right and A^{-1} on the left,

$$A^{-1} \hat{F}_A (\hat{F}_A' A \bar{\mathcal{F}}) = \bar{\mathcal{F}} + \alpha, \qquad \alpha = \frac{1}{n} \sum_{i=1}^{r} A^{-1} P_{Ai}^{(1)} A \bar{\mathcal{F}} + A^{-1} \delta_{A1} A \bar{\mathcal{F}},$$

so $P_{\tilde{F}}$ is the projection onto the span of $\bar{\mathcal{F}} + \alpha$. With $\|\alpha\| = O_P(n^{-1})$, $\bar{\mathcal{F}}'\bar{\mathcal{F}} = I$ and $\bar{\mathcal{F}}\bar{\mathcal{F}}' = P_0$,

$$\left[(\bar{\mathcal{F}} + \alpha)'(\bar{\mathcal{F}} + \alpha)\right]^{-1} = I - [\alpha'\bar{\mathcal{F}} + \bar{\mathcal{F}}'\alpha] + O_P(n^{-2}),$$

and therefore

$$P_{\tilde{F}} = P_0 + M_0\alpha\bar{\mathcal{F}}' + \bar{\mathcal{F}}\alpha' M_0 + O_P(n^{-2}). \tag{A.4}$$

Taking the trace of the product of (A.1) and (A.4), and using $\operatorname{tr}[P_0 P_i^{(1)}] = 0$ and $P_0 M_0 = 0$, gives $\operatorname{tr}[P_{\hat{F}} P_{\tilde{F}}] = r + O_P(n^{-2})$, i.e. $S = O_P(n^{-2})$.

Remark A.1 (Robustness to clustered factor eigenvalues)**.** The individual reduced resolvents S_i carry within-block terms $(\mu_j - \mu_i)^{-1}$, $i, j \le r$, which diverge when the factor eigenvalues cluster. These terms never enter the result, because they cancel in every block sum. At first order, the within-block part of $\sum_{i \le r} P_i^{(1)}$ is

$$-\sum_{i \neq j} \frac{P_i \Phi_1 P_j + P_j \Phi_1 P_i}{\mu_j - \mu_i} = 0,$$

since the (i, j) and (j, i) contributions carry the opposite coefficients $(\mu_j - \mu_i)^{-1}$ and $(\mu_i - \mu_j)^{-1}$; only the block-to-noise part $\sum_i \mu_i^{-1} P_i \Phi_1 M_0 = \Phi_0^+ \Phi_1 M_0$ survives. At second order, $\operatorname{tr}[P_0 P_i^{(2)}] = -\mu_i^{-2} \operatorname{tr}[P_i \Phi_1 M_0 \Phi_1 P_i]$ contains no within-block difference at all. Equivalently, P_0, Φ_0^+, D_A and the scalar S are functions of the rank-r *group* projector alone (Appendix E), so the expansion and its remainders $\delta_1 = O_P(n^{-2})$, $\delta_2 = O_P(n^{-3})$ are governed by the block-to-noise gap $\mu_r = \Theta_P(1)$ (Assumption **A1**), and hold uniformly whether or not the factor eigenvalues are separated. This is what licenses the clustered-eigenvalue regime of Section 3.2, in which individual eigenvectors are inconsistent yet the block statistic is not.

A.2 Second-order expansion and the reduction

Expanding (A.1) one order further,

$$P_{\hat{F}} = P_0 + \frac{1}{n}\sum_{i=1}^{r} P_i^{(1)} + \frac{1}{n^2}\sum_{i=1}^{r} P_i^{(2)} + \delta_2, \qquad \|\delta_2\| = O_P(n^{-3}), \tag{A.5}$$

where, following Kato (1980, pp. 75–77),

$$\begin{aligned} P_i^{(2)} = {} & P_i\Phi_1 S_i \Phi_1 S_i + S_i \Phi_1 P_i \Phi_1 S_i + S_i \Phi_1 S_i \Phi_1 P_i \\ & - P_i \Phi_1 P_i \Phi_1 S_i^2 - P_i \Phi_1 S_i^2 \Phi_1 P_i - S_i^2 \Phi_1 P_i \Phi_1 P_i, \end{aligned}$$

with $\left\|\sum_i P_i^{(2)}\right\| = O_P(1)$, and similarly for $P_{\hat{F}_A}$. Proceeding as above,

$$P_{\tilde{F}} = (\bar{\mathcal{F}} + \beta)\left[(\bar{\mathcal{F}} + \beta)'(\bar{\mathcal{F}} + \beta)\right]^{-1}(\bar{\mathcal{F}} + \beta)',$$
$$\beta = \frac{1}{n}\sum_i A^{-1}P_{Ai}^{(1)}A\bar{\mathcal{F}} + \frac{1}{n^2}\sum_i A^{-1}P_{Ai}^{(2)}A\bar{\mathcal{F}} + A^{-1}\delta_{A2}A\bar{\mathcal{F}},$$

and expanding the inverse to second order yields

$$\begin{aligned} P_{\tilde{F}} = {} & P_0 + \bar{\mathcal{F}}\beta' M_0 + M_0\beta\bar{\mathcal{F}}' + M_0\beta\beta' M_0 \\ & - (\bar{\mathcal{F}}\beta')^2 M_0 - M_0(\beta\bar{\mathcal{F}}')^2 - \bar{\mathcal{F}}\beta' M_0\beta\bar{\mathcal{F}}' + O_P(n^{-3}). \end{aligned} \tag{A.6}$$

Taking the trace of the product of (A.5) and (A.6),

$$\operatorname{tr}[P_{\hat{F}}P_{\tilde{F}}] = r - \operatorname{tr}[\beta' M_0\beta] + \frac{2}{n}\sum_i \operatorname{tr}[P_i^{(1)}M_0\beta\bar{\mathcal{F}}'] + \frac{1}{n^2}\sum_i \operatorname{tr}[P_0 P_i^{(2)}] + O_P(n^{-3}).$$

Using $\sum_i P_i^{(1)}M_0 = \Phi_0^+\Phi_1 M_0$ and $\operatorname{tr}[P_0 P_i^{(2)}] = -\mu_i^{-2}\operatorname{tr}[P_i\Phi_1 M_0\Phi_1 P_i]$, so that $\sum_i \operatorname{tr}[P_0 P_i^{(2)}] = -\operatorname{tr}[\Phi_0^+\Phi_1 M_0\Phi_1\Phi_0^+]$, and replacing β by its leading term (the difference is $O_P(n^{-3})$),

$$\begin{aligned} n^2 S = {} & \sum_{i,j=1}^{r} \operatorname{tr}\left[A' P_{Ai}^{(1)} A'^{-1} M_0 A^{-1} P_{Aj}^{(1)} A P_0\right] \\ & - 2\sum_{j=1}^{r} \operatorname{tr}[\Phi_0^+\Phi_1 M_0 A^{-1} P_{Aj}^{(1)} A] + \operatorname{tr}[\Phi_0^+\Phi_1 M_0\Phi_1\Phi_0^+] + O_P(n^{-1}). \end{aligned} \tag{A.7}$$

The first sum simplifies through the identity

$$M_0 A^{-1} P_{Aj} = 0, \tag{A.8}$$

valid because P_{Aj} maps into the span of $A\mathcal{F}$, so $A^{-1}P_{Aj}$ maps into the span of $\mathcal{F}$, which M_0 annihilates. With (A.8), (A.3), $\sum_j \mu_{Aj}^{-1} P_{Aj} = \Phi_{A0}^+$, $M_{A0}A'^{-1}M_0 A^{-1}M_{A0} = A'^{-1}M_0 A^{-1}$ and $\Phi_{A1} = A\Phi_1 A'$,

$$\sum_{i,j} \operatorname{tr}[A' P_{Ai}^{(1)} A'^{-1} M_0 A^{-1} P_{Aj}^{(1)} A P_0] = \operatorname{tr}[A'\Phi_{A0}^+ A\Phi_1 M_0\Phi_1 A'\Phi_{A0}^+ A P_0],$$

and likewise $\sum_j \operatorname{tr}[\Phi_0^+ \Phi_1 M_0 A^{-1} P_{Aj}^{(1)} A] = \operatorname{tr}[\Phi_0^+ \Phi_1 M_0 \Phi_1 A' \Phi_{A0}^+ A]$. Substituting and using $\Phi_0^+ = P_0 \Phi_0^+$,

$$n^2 S = \operatorname{tr}[P_0 D_A \Phi_1 M_0 \Phi_1 D_A P_0] + O_P(n^{-1}) = \|P_0 D_A \Phi_1 M_0\|_F^2 + O_P(n^{-1}),$$
$$D_A \equiv A' \Phi_{A0}^+ A - \Phi_0^+,$$

which is Theorem 2.1. □

B Proof of Theorem 3.2

We prove the one-factor limiting distribution; throughout $r = 1$ and the factor index is suppressed, with limits as $N, T \to \infty$, $T/N \to c$.

B.1 Reduction to quadratic forms in f

Recall the notation from Section 2. Theorem 2.1 gives

$$n^2 S = \|P_0 D_A \Phi_1 M_0\|_F^2 + O_p(n^{-1}), \tag{B.1}$$

where $\Phi_1 = e M_\lambda e'/\lambda_2 = (N/\lambda_2)\Delta$ and $P_0 = ff'/f'f$ (the projection onto the one-dimensional factor space).

For $r = 1$ the Moore-Penrose pseudoinverse satisfies $\Phi_0^+ = (\lambda_1/\|F\|^2)(ff'/f'f)$, and an analogous formula holds for Φ_{A0}^+. Writing $W = A'A$ and using $\lambda_1/\|F\|^2 = 1 + O_p(n^{-1})$ (Weyl's inequality), straightforward algebra gives

$$n^2 S = \left(\frac{N}{\lambda_2}\right)^2 \left[\frac{Q_1 Q_7}{Q_2^2} - 2\frac{Q_6}{Q_2} + \frac{Q_5}{Q_1} - \left(\frac{Q_4}{Q_2} - \frac{Q_3}{Q_1}\right)^2\right] + O_p(n^{-1}), \tag{B.2}$$

where the seven quadratic forms in f are

	Q_1	Q_2	Q_3	Q_4	Q_5	Q_6	Q_7
def.:	$f'f$	$f'Wf$	$f'\Delta f$	$f'W\Delta f$	$f'\Delta^2 f$	$f'W\Delta^2 f$	$f'W\Delta^2 W f$.

(B.3)

(See the derivation in the Online Appendix for the complete algebra.)

B.2 Probability limits of the quadratic forms

Since $f = \Sigma_F^{1/2}\xi/\sqrt{T}$ and $\xi \sim N(0, I_T)$ is independent of Δ, the law of large numbers for quadratic forms gives $f'Vf - T^{-1}\operatorname{tr}(V) \xrightarrow{p} 0$ for any matrix V with $\operatorname{tr}(V^2)/T^2 \to 0$. Combined with freeness of Δ^k and W^j (Anderson et al. (2009), Corollary 5.4.11), which gives

$$\frac{1}{T}\operatorname{tr}(\Delta^k W^j) - \frac{1}{T}\operatorname{tr}(\Delta^k)\cdot\frac{1}{T}\operatorname{tr}(W^j) \xrightarrow{p} 0,$$

we obtain

	Q_1	Q_2	Q_3	Q_4	Q_5	Q_6	Q_7
plim :	1	τ_1	ρ_1	$\tau_1\rho_1$	ρ_2	$\tau_1\rho_2$	$\tau_2\rho_2$.

(B.4)

Denote the limiting value vector A_j for Q_j. Using these limits in (B.2) and recalling $(\lambda_1/N)^2 = (N/\lambda_2)^2\cdot(\lambda_1/\lambda_2)^{-2} = (N/\lambda_2)^2\cdot n^{-2}$ we obtain

$$\left(\frac{\lambda_1}{N}\right)^2 S \xrightarrow{p} \rho_2\left(\frac{\tau_2}{\tau_1^2} - 1\right) = \rho_2(\gamma_2 - 1) \equiv \omega_1. \tag{B.5}$$

B.3 Central limit theorem for the quadratic forms

By the order lemma proved in the Online Appendix, $Q_j = A_j + O_p(T^{-1/2})$ for each j. The delta method applied to the twice-differentiable function $g(q_1, \ldots, q_7) = q_1q_7/q_2^2 - 2q_6/q_2 + q_5/q_1 - (q_4/q_2 - q_3/q_1)^2$ gives

$$\sqrt{T}[g(Q) - g(A)] = \sum_{j=1}^{7} w_j\,\sqrt{T}(Q_j - A_j) + O_p(T^{-1/2} + T^{1/2}n^{-1}), \tag{B.6}$$

where the partial derivatives at A are

$$w_1 = \omega_1, \quad w_2 = -2\omega_1/\tau_1, \quad w_3 = w_4 = 0,$$
$$w_5 = 1, \quad w_6 = -2/\tau_1, \quad w_7 = 1/\tau_1^2.$$

(The vanishing of w_3 and w_4 follows from a direct computation of $\partial g/\partial q_3$ and $\partial g/\partial q_4$ evaluated at A.) Under the condition $T^{1/2}n^{-1} \to 0$ the remainder is $o_p(1)$ after multiplication by $\sqrt{T}$.

Since $\xi \sim N(0, I_T)$, the linear combination $\sum_j w_j\sqrt{T}(Q_j - A_j)$ can be written as

$\sqrt{T}(\xi' B_T \xi/T - T^{-1}\operatorname{tr} B_T)$, where

$$B_T = w_1\Sigma_F - \frac{2w_1}{\tau_1}\Sigma_F^{1/2} W \Sigma_F^{1/2} + \Sigma_F^{1/2}\Delta^2\Sigma_F^{1/2} - \frac{2}{\tau_1}\Sigma_F^{1/2} W \Delta^2 \Sigma_F^{1/2} + \frac{1}{\tau_1^2}\Sigma_F^{1/2} W \Delta^2 W \Sigma_F^{1/2}. \tag{B.7}$$

The CLT (3.6) then follows from the quadratic-form CLT for independent entries (Bai and Silverstein, 1998; Capitaine et al., 2009). Its conditions hold for the B_T of (B.7): $\|B_T\| = O_P(1)$ (bounded spectral norms of Σ_F, W, Δ under **A3**,**B2**, and $w_j = O_P(1)$) while $\operatorname{tr} B_T^2 = \Theta_P(T)$ (as $\Omega = 2\operatorname{plim} T^{-1}\operatorname{tr} B_s^2 > 0$), so $\|B_T\|^2/\operatorname{tr} B_T^2 = O_P(1/T) \to 0$ (the Lindeberg-type condition), and $\omega = \operatorname{plim} \sum_i B_{s,ii}^2/\operatorname{tr} B_T^2$ converges—the delocalization functional of the non-Gaussian correction. The trace evaluations are in the Online Appendix.

B.4 The asymptotic variance

The variance of the Gaussian quadratic form depends on B_T only through its symmetric part $B_s = \frac{1}{2}(B_T + B_T')$, with $\operatorname{Var}(\xi' B_s \xi) = 2\operatorname{tr} B_s^2$; hence

$$\Omega = 2\operatorname{plim}_{N,T\to\infty} T^{-1}\operatorname{tr} B_s^2.$$

Substituting (B.7) and taking probability limits of the trace functionals ρ_2, ρ_4 of Δ and τ_k, τ_{ab} of Σ_F, W yields the explicit form Ω of (3.8); the evaluation is a routine but lengthy expansion of $\operatorname{tr} B_s^2$ into these nine functionals, carried out term by term in the Online Appendix. Under non-Gaussian factors the identity acquires the extra term $\kappa_F \sum_i B_{s,ii}^2$, giving the non-Gaussian correction $\Omega(\kappa_F) = \Omega + \kappa_F \Psi$ (Online Appendix). This proves Theorem 3.2. □

C Proof of Lemma 3.1

Proof. Immediate: $n \asymp_P N^{\alpha_r - \alpha_e}$ and $T^{1/2} = N^{(\ln T)/(2\ln N)}$, so $n \gg T^{1/2} \iff N^{\alpha_r-\alpha_e} \gg N^{(\ln T)/(2\ln N)} \iff \alpha_r - \alpha_e > (\ln T)/(2\ln N)$. □

D Proof of Proposition 3.3

Proof. Write $Q = (\lambda_1/N)^2 S$. By (3.9), $Q - \hat{\omega}_1 \xrightarrow{P} q_* - \omega_* > 0$ and $\hat{\Omega} \xrightarrow{P} \Omega_* \in (0, \infty)$, so $(Q - \hat{\omega}_1)/\sqrt{\hat{\Omega}} \xrightarrow{P} (q_* - \omega_*)/\sqrt{\Omega_*} > 0$. Hence $Z = \sqrt{T}\big[(q_* - \omega_*)/\sqrt{\Omega_*} + o_P(1)\big] \xrightarrow{P} +\infty$, and since the critical value $z_{1-\alpha}$ is fixed, $\Pr(Z > z_{1-\alpha}) \to 1$. For the omnibus statistic the

argument applies to each per-direction summand; a maximum over a fixed finite library of individually diverging statistics diverges, giving consistency of $\max_{A\in\mathcal{L}} Z_{\mathrm{blk}}(A)$. □

E The multi-factor extension

We prove Theorems 3.4, 3.6, 3.7, Lemma 3.5 and Proposition 3.9, deferring the computations to the Online Appendix (Appendix C). All limits are as $N, T \to \infty$, $T/N \to c$, on $\{\hat{r} = r\}$, conditioning on ε (where $e = \varepsilon\Sigma^{1/2}$) for the distributional steps. The proof runs on one mechanism: the block statistic is a smooth functional of $r \times r$ moment matrices with *diagonal* limits, whose differential annihilates the off-diagonal fluctuations, so it linearises into a sum of one-factor statistics in the *independent* $\xi_1, \ldots, \xi_r$.

Block reduction (Theorem 3.4). With the rank-one eigenprojector of Appendix B replaced by the rank-r Kato group projector $P_0 = \sum_{i\le r} \pi_i$, the residue calculus cancels the within-block eigenvalue differences, so only the block gap $n = \lambda_r/\lambda_{r+1}$ enters and Appendix A carries over to $n^2 S = \|P_0 D_A \Phi_1 M_0\|_F^2 + O_P(n^{-1})$, with $D_A = A'\Phi_{A0}^{+}A - \Phi_0^{+}$ and $\Phi_0^{+} = \bar{F}\Lambda_{\mathrm{blk}}\bar{F}'/T^2$, uniformly in the within-block spacings.

Moment matrices and decoupling. Under **B1′**, write $g_a = \Sigma_{F,a}^{1/2}\xi_a$, $G = (g_1, \ldots, g_r)$, $\bar{F} = G(G'G/T)^{-1/2}$, and, for the kernels $V_1 = I, V_2 = W, V_3 = \Delta, V_4 = W\Delta, V_5 = \Delta^2, V_6 = W\Delta^2, V_7 = W\Delta^2 W$, the moment matrices $\widehat{\mathcal{Q}}_j = T^{-1}G'V_jG$ and $\mathcal{Q}_j = T^{-1}\bar{F}'V_j\bar{F}$ (so $\mathcal{Q}_1 = I_r$). Two lemmas, proved in the Online Appendix, drive the reduction.

Lemma E.1 (Diagonal limits; single-index fluctuations). $\widehat{\mathcal{Q}}_j \xrightarrow{p} \bar{\mathcal{Q}}_j = \mathrm{diag}(\bar{q}_j^{(a)})_a$ *with* $(\bar{q}_1, \ldots, \bar{q}_7)^{(a)} = (1, \tau_1^{(a)}, \rho_1, \tau_1^{(a)}\rho_1, \rho_2, \tau_1^{(a)}\rho_2, \tau_2^{(a)}\rho_2)$, $\tau_k^{(a)} = \mathrm{plim} T^{-1}\,\mathrm{tr}(\Sigma_{F,a}W^k)$*; and, up to* $o_p(T^{-1/2})$*, the diagonal fluctuation* $(\mathcal{Q}_j)_{aa} - \bar{q}_j^{(a)}$ *reduces to a quadratic form in* ξ_a alone *(the orthonormaliser couples directions only at negligible order; Online Appendix).*

By Theorem 3.4 and the algebra of Appendix B carried on the $\mathcal{Q}_j$, with $D := \Lambda_{\mathrm{blk}}/\lambda_r$ and $M := D\mathcal{Q}_2^{-1}\mathcal{Q}_4 - D\mathcal{Q}_3$,

$$\left(\tfrac{\lambda_r}{N}\right)^2 S = \Psi_\Lambda(\mathcal{Q}) + O_p(n^{-1}), \quad \Psi_\Lambda = \mathrm{tr}\left[D\mathcal{Q}_2^{-1}\mathcal{Q}_7\mathcal{Q}_2^{-1}D - 2D\mathcal{Q}_2^{-1}\mathcal{Q}_6 D + D\mathcal{Q}_5 D - MM'\right],$$

the matrix analogue of the one-factor bracket g of Appendix B.

Lemma E.2 (Off-diagonal annihilation). $\bar{M} := D(\bar{\mathcal{Q}}_2^{-1}\bar{\mathcal{Q}}_4 - \bar{\mathcal{Q}}_3) = 0$*, and the differential of* Ψ_Λ *at the diagonal limit depends only on the diagonal perturbations:* $D\Psi_\Lambda[\delta\mathcal{Q}] = \sum_a \sum_j w_j^{(a)}(\delta\mathcal{Q}_j)_{aa}$ *with* $w_3^{(a)} = w_4^{(a)} = 0$ *and* $w_j^{(a)} = (\lambda_a/\lambda_r)^2\,\partial g/\partial q_j$ *the one-factor weights at* $\Sigma_{F,a}$.

A second-order Taylor expansion of Ψ_Λ—its quadratic term $O_p(T^{-1})$ being the cross-direction remainder $\mathcal{E}_T$ of (3.10), so $\sqrt{T}(\lambda_r/N)^2\mathcal{E}_T = o_p(1)$ (Lemma 3.5(ii))—with Lemmas E.1–E.2 gives

$$\sqrt{T}\Big[\big(\tfrac{\lambda_r}{N}\big)^2 S - \textstyle\sum_a \big(\tfrac{\lambda_r}{\lambda_a}\big)^2 \omega_1^{(a)}\Big] = \textstyle\sum_{a=1}^r \big(\tfrac{\lambda_r}{\lambda_a}\big)^2 U_a + o_p(1), \tag{E.1}$$

where $U_a := \sqrt{T}[(\lambda_a/N)^2 S_a - \omega_1^{(a)}]$ is *verbatim* the one-factor statistic of Appendix B at $\Sigma_{F,a}$, a function of (ξ_a, ε) alone; this is Lemma 3.5(i)–(ii).

Joint limit and independence (Lemma 3.5(iii); Theorems 3.6–3.7). Conditional on ε, each U_a is a function of ξ_a alone (Lemma E.1) and the ξ_a are independent (**B1′**), so the U_a are conditionally independent—*derived, not assumed*—and each obeys the one-factor CLT of Appendix B; a characteristic-function argument (Online Appendix) gives $(U_1, \ldots, U_r) \xrightarrow{d} N(0, \mathrm{diag}(\Omega^{(a)}))$, joint normality *and* asymptotic independence. The weighted sum (E.1) yields Theorem 3.7; Case H (Theorem 3.6) is the sub-case $\Sigma_{F,a} \equiv \Sigma_F$, and $Z_{\mathrm{blk}} \xrightarrow{d} N(0,1)$ by Slutsky.

Identification boundary (Proposition 3.9). By (3.8), $\sum_a \Omega^{(a)}$ depends on $\sum_a \mathrm{tr}(\Sigma_{F,a} W \Sigma_{F,a} W)$, not on the block sum $\bar{\Sigma} = \sum_a \Sigma_{F,a}$ (an explicit counterexample is in the Online Appendix); under clustered eigenvalues only $\bar{\Sigma}$-measurable quantities are estimable, so only Case H is feasible, while the separated regime identifies each $\Sigma_{F,a}$ and gives Case K. □

Supplementary Material

Supplement to "Factor Dipping"

F Supporting computations for Theorem 3.2

This appendix supplies the lengthy computations deferred from the proof of Theorem 3.2 in Appendix B of the main text: the evaluation of the asymptotic variance Ω, the order lemma, and the derivation of the quadratic-form representation. Notation is as in Appendix B.

Throughout this appendix we work with $r = 1$ and suppress the factor index. All limits are as $N, T \to \infty$ with $T/N \to c$.

F.1 Computation of the asymptotic variance Ω

The asymptotic variance of the quadratic form $\xi' B_T \xi$ depends on B_T only through its symmetric part $B_s = \frac{1}{2}(B_T + B_T')$, and $\mathrm{Var}(\xi' B_s \xi) = 2 \operatorname{tr} B_s^2$; hence $\Omega = 2 \operatorname{plim} T^{-1} \operatorname{tr} B_s^2$.

Non-Gaussian factors. The identity $\mathrm{Var}(\xi' B_s \xi) = 2 \operatorname{tr} B_s^2$ uses the Gaussianity of ξ (Assumption **B1**). For ξ with i.i.d. entries of mean 0, variance 1 and finite fourth moment, with excess kurtosis $\kappa_F = \mathbb{E}[\xi_1^4] - 3$, the standard quadratic-form variance identity reads

$$\mathrm{Var}(\xi' B_s \xi) = 2 \operatorname{tr} B_s^2 + \kappa_F \sum_{i=1}^{T} B_{s,ii}^2,$$

the second term vanishing when ξ is Gaussian ($\kappa_F = 0$). Hence, under the relaxed **B1**, the asymptotic variance in Theorem 3.2 becomes

$$\Omega(\kappa_F) = \Omega + \kappa_F \, \Psi, \qquad \Psi = \operatorname{plim}_{N,T\to\infty} \frac{1}{T} \sum_{i=1}^{T} B_{s,ii}^2 \;\geq\; 0, \tag{F.1}$$

with Ω the Gaussian expression (3.8) and $B_{s,ii}$ the diagonal entries, in the time basis in which ξ has i.i.d. components, of the symmetric matrix B_s in (B.7). The idiosyncratic matrix enters B_T only through the concentrated spectral moments ρ_2, ρ_4 of Δ, whose fluctuations about their probability limits are $O_p\big((NT)^{-1/2}\big)$ and hence negligible after the $\sqrt{T}$ scaling; the idiosyncratic distribution therefore does not affect Ω, and Assumption **B2** may be weakened to i.i.d. mean-zero, unit-variance entries with finite fourth moment without changing the limit. Both κ_F and Ψ are consistently estimable: $\hat{\kappa}_F$ by the sample excess kurtosis of the

estimated factor innovations $\hat{\Sigma}_F^{-1/2}\hat{F}$, and $\hat{\Psi} = T^{-1}\sum_i \hat{B}_{s,ii}^2$ by plugging $\hat{\Sigma}_F, \hat{W}, \hat{\Delta}$ into (B.7). This is collected as Proposition G.4 in the main text.

Every term of (B.7) is symmetric except the cross term $-(2/\tau_1)\Sigma_F^{1/2}W\Delta^2\Sigma_F^{1/2}$, whose symmetric form is $-(1/\tau_1)\Sigma_F^{1/2}\big(W\Delta^2 + \Delta^2 W\big)\Sigma_F^{1/2}$. This is a genuine correction: for any real matrix $\operatorname{tr} B_T^2 = \operatorname{tr} B_s^2 - \|B_a\|_F^2$ with B_a the antisymmetric part, so $2\operatorname{tr} B_T^2$ understates the variance (and can be negative); the deficit is concentrated in this single term and equals $4(\gamma_2 - 1)(\rho_4 - \rho_2^2)$ in the probability limit. Expanding $\operatorname{tr} B_s^2$ (only the $W\Delta^2$ self-term differs from $\operatorname{tr} B_T^2$, where it enters symmetrized):

$$\begin{aligned}
\operatorname{tr} B_T^2 = {} & w_1^2 \operatorname{tr}\Sigma_F^2 + \frac{4w_1^2}{\tau_1^2}\operatorname{tr}(\Sigma_F W)^2 + \operatorname{tr}(\Sigma_F\Delta^2)^2 \\
& + \frac{2}{\tau_1^2}\big[\operatorname{tr}(\Sigma_F W\Delta^2)^2 + \operatorname{tr}(\Sigma_F W\Delta^2\Sigma_F\Delta^2 W)\big] + \frac{1}{\tau_1^4}\operatorname{tr}(W\Sigma_F W\Delta^2)^2 \\
& - \frac{4w_1^2}{\tau_1}\operatorname{tr}[\Sigma_F^2 W] + 2w_1\operatorname{tr}[\Sigma_F^2\Delta^2] - \frac{4w_1}{\tau_1}\operatorname{tr}\Sigma_F^2 W\Delta^2 + \frac{2w_1}{\tau_1^2}\operatorname{tr} W\Sigma_F^2 W\Delta^2 \\
& - \frac{4w_1}{\tau_1}\operatorname{tr}[\Sigma_F W\Sigma_F\Delta^2] + \frac{8w_1}{\tau_1^2}\operatorname{tr}[(\Sigma_F W)^2\Delta^2] - \frac{4w_1}{\tau_1^3}\operatorname{tr} W(\Sigma_F W)^2\Delta^2 \\
& - \frac{4}{\tau_1}\operatorname{tr} W(\Delta^2\Sigma_F)^2 + \frac{2}{\tau_1^2}\operatorname{tr}[\Sigma_F\Delta^2\Sigma_F W\Delta^2 W] - \frac{4}{\tau_1^3}\operatorname{tr} W(\Sigma_F W\Delta^2)^2. \qquad \text{(F.2)}
\end{aligned}$$

To evaluate the probability limit of each term, we use the free probability formula for mixed moments of freely independent matrices (Anderson et al. 2009, eq. (5.3.20)–(5.3.21)): for freely independent $a, c \in \mathcal{A}_1$ and $b \in \mathcal{A}_2$,

$$\varphi(abcb) = \varphi(a)\varphi(c)\big(\varphi(b^2) - \varphi(b)^2\big) + \varphi(ac)\varphi(b)^2, \qquad \text{(F.3)}$$

where $\varphi(\cdot) = \operatorname{plim} T^{-1}\operatorname{tr}(\cdot)$. In our setting, Σ_F and W belong to one sub-algebra while Δ belongs to a freely independent sub-algebra (by the random matrix freeness results cited above). This gives, e.g.,

$$\operatorname{plim}\frac{1}{T}\operatorname{tr}(\Sigma_F W\Delta^2)^2 = \tau_1^2(\rho_4 - \rho_2^2) + \tau_{11}\rho_2^2,$$

and similarly for every other trace in (F.2). (The complete list of free-probability evaluations is carried out term by term below.)

Evaluating all terms and using $w_1 = \rho_2(\gamma_2 - 1)$, then collecting and simplifying with the γ and γ_{ab} notation, yields

$$\operatorname{plim}\frac{\operatorname{tr} B_T^2}{T} = \big(\gamma_{22} - 4\gamma_2\gamma_{12} + 4\gamma_2^2\gamma_{11} + \gamma_2^2(\gamma_{00} - 4\gamma_{01}) + 2\gamma_2\gamma_{02}\big)\,\rho_2^2 + (\gamma_2 - 1)^2(\rho_4 - \rho_2^2), \qquad \text{(F.4)}$$

where the $(\rho_4 - \rho_2^2)$ coefficient $(\gamma_2 - 1)^2 = (\gamma_2^2 - 4\gamma_2 + 3) + 2(\gamma_2 - 1)$ already incorporates the $+2(\gamma_2 - 1)$ contributed by the symmetrization of the $W\Delta^2$ term above. Multiplying by 2 gives Ω as stated in (3.8); the coefficient $2(\gamma_2 - 1)^2 \geq 0$ is a perfect square, as a variance component multiplying $\rho_4 - \rho_2^2 \geq 0$ must be.

F.1.1 Detailed free-probability evaluations

We now carry out each term of (F.2). All probability limits are taken as $N, T \to \infty$ and we write $\varphi(V) = \text{plim} T^{-1} \operatorname{tr} V$.

Pure Σ_F–W terms (no Δ). These use only $\varphi(\Sigma_F^a W^b) = \tau_{0b} \cdot$ [normalization] and the definitions of τ_k, τ_{ab}:

$$\begin{aligned}
\varphi(\Sigma_F^2) &= \tau_{00}, \\
\varphi((\Sigma_F W)^2) &= \tau_{11}, \\
\varphi(\Sigma_F^2 W) &= \tau_{01}, \\
\varphi(W\Sigma_F^2 W) &= \tau_{02} \text{ (by cyclic trace property).}
\end{aligned}$$

Terms mixing Δ^2 with Σ_F–W products. Apply (F.3) with $\varphi(\Delta^2) = \rho_2$ and $\varphi(\Delta^4) \equiv \rho_4$:

$$\begin{aligned}
\varphi((\Sigma_F\Delta^2)^2) &= \tau_0^2(\rho_4 - \rho_2^2) + \tau_{00}\rho_2^2 = (\rho_4 - \rho_2^2) + \tau_{00}\rho_2^2 \qquad (\tau_0 = \varphi(\Sigma_F) = 1), \\
\varphi((\Sigma_F W\Delta^2)^2) &= \tau_1^2(\rho_4 - \rho_2^2) + \tau_{11}\rho_2^2, \\
\varphi(\Sigma_F W\Delta^2\Sigma_F\Delta^2 W) &= \tau_2(\rho_4 - \rho_2^2) + \tau_{11}\rho_2^2 \qquad \text{(symmetrization partner of the } W\Delta^2 \text{ term)}, \\
\varphi((W\Sigma_F W\Delta^2)^2) &= \tau_2^2(\rho_4 - \rho_2^2) + \tau_{22}\rho_2^2, \\
\varphi(\Sigma_F^2\Delta^2) &= \tau_{00}\rho_2, \\
\varphi(\Sigma_F^2 W\Delta^2) &= \tau_{01}\rho_2, \\
\varphi(W\Sigma_F^2 W\Delta^2) &= \tau_{02}\rho_2, \\
\varphi(\Sigma_F W\Sigma_F\Delta^2) &= \tau_{01}\rho_2, \\
\varphi((\Sigma_F W)^2\Delta^2) &= \tau_{11}\rho_2.
\end{aligned}$$

Higher-order mixed terms. For terms of the form $W(\Sigma_F W)^2\Delta^2$ we apply Anderson et al. (2009), eq. (5.3.21):

$$\begin{aligned}
&\varphi(W(\Sigma_F W)^2\Delta^2) = (\tau_3 - \tau_1\tau_2)\rho_2 \cdot \frac{1}{\tau_1} + \tau_{12}\rho_2 \cdot \frac{1}{\tau_1} \\
\Longrightarrow \quad &\varphi(W(\Sigma_F W)^2\Delta^2) = \tau_{12}\rho_2 + (\tau_3 - \tau_1\tau_2)\rho_2/\tau_1.
\end{aligned}$$

More precisely, using the notation $\tau_{ab} = \text{plim} T^{-1}\operatorname{tr}(\Sigma_F W^a \Sigma_F W^b)$, the relevant evaluations are

$$\begin{aligned}\varphi(W(\Delta^2\Sigma_F)^2) &= \tau_{11}(\rho_4 - \rho_2^2) + \tau_{01}\rho_2^2,\\ \varphi(\Sigma_F\Delta^2\Sigma_F W\Delta^2 W) &= \tau_1^2(\rho_4 - \rho_2^2) + \tau_{02}\rho_2^2,\\ \varphi(W(\Sigma_F W\Delta^2)^2) &= \tau_2\tau_1(\rho_4 - \rho_2^2) + \tau_{12}\rho_2^2.\end{aligned}$$

These follow from applying (F.3) iteratively, treating the Σ_F–W cluster as element a or c and Δ^2 as element b.

Substituting all the above into (F.2), collecting terms in ρ_2^2, $\rho_4 - \rho_2^2$, and ρ_4, and recalling $w_1 = \rho_2(\gamma_2 - 1)$, $\tau_k/\tau_1^k = \gamma_k$, $\tau_{ab}/\tau_1^{a+b} = \gamma_{ab}$, one arrives after elementary algebra at (F.4). The full algebraic verification is included in the supplementary computation file.

F.2 Proof of Lemma F.1

Lemma F.1. *Under Assumptions A1–A3, B1–B4, $Q_j = A_j + O_p(T^{-1/2})$ for each $j \in \{1, \dots, 7\}$.*

Proof. By Remark F.2 (a special case of the quadratic-form CLT), for any deterministic matrix V with $\operatorname{tr} V^2 = O(T)$,

$$f'Vf - \frac{1}{T}\operatorname{tr}(\Sigma_F V) = O_p(T^{-1/2}).$$

We apply this to each Q_j. For example, $Q_1 = f'f$, and $T^{-1}\operatorname{tr}(\Sigma_F I_T) = T^{-1}\operatorname{tr}\Sigma_F = 1 = A_1$. For $Q_5 = f'\Delta^2 f$, we condition on Δ (which is independent of ξ) and apply the LLN conditioned on Δ: $Q_5 - T^{-1}\operatorname{tr}(\Sigma_F\Delta^2) = O_p(T^{-1/2})$, and $T^{-1}\operatorname{tr}(\Sigma_F\Delta^2) \xrightarrow{p} \rho_2 = A_5$ by freeness and the Marchenko-Pastur limit for Δ. The other Q_j follow identically. □

Remark F.2. As a special case of the quadratic-form CLT (with $B = V/\sqrt{\operatorname{tr} V^2}$ and checking Assumption (i)), if $\|V\|^2/\operatorname{tr} V^2 \to 0$ in probability then $(Y_T'VY_T - \operatorname{tr} V)/\sqrt{2\operatorname{tr} V^2} \xrightarrow{d} N(0,1)$. In particular, if $\operatorname{tr} V^2 \to 0$ then $Y_T'VY_T - \operatorname{tr} V \xrightarrow{p} 0$.

F.3 Derivation of the quadratic form representation (B.2)

Starting from Theorem 2.1, $n^2 S = \|P_0 D_A \Phi_1 M_0\|_F^2 + O_p(n^{-1})$. For $r = 1$,

$$P_0 = \frac{ff'}{f'f}, \quad \Phi_0^+ = \frac{\lambda_1}{\|F\|^2}P_0, \quad A'\Phi_{A0}^+ A = \frac{\lambda_1}{\|F\|^2}\cdot\frac{Wff'W}{f'Wf},$$

so

$$D_A = A'\Phi_{A0}^{+}A - \Phi_0^{+} = \frac{\lambda_1}{\|F\|^2}\left(\frac{Wff'W}{f'Wf} - \frac{ff'}{f'f}\right).$$

Using $\lambda_1/\|F\|^2 = 1 + O_p(n^{-1})$ and $\Phi_1 = (N/\lambda_2)\Delta$,

$$n^2 S = \left\| P_0\left(\frac{Wff'W}{f'Wf} - \frac{ff'}{f'f}\right)\Delta M_0 \right\|_F^2 \left(\frac{N}{\lambda_2}\right)^2 + O_p(n^{-1}).$$

Because $P_0 = ff'/(f'f)$ and $P_0 M_0 = 0$, and writing $P_0 W = ff'W/(f'f)$, $P_0 Wff'W/(f'Wf) = ff'Wff'W/(f'f \cdot f'Wf)$, expanding $\|P_0(\cdots)\Delta M_0\|_F^2 = \mathrm{tr}[P_0 D_A \Delta M_0 \Delta D_A P_0]$ and using the cyclic trace property yields

$$\|P_0 D_A \Delta M_0\|_F^2 = \frac{f'W\Delta^2 Wf}{f'Wf} - 2\frac{f'W\Delta^2 f}{f'Wf} + \frac{f'\Delta^2 f}{f'f} - \left(\frac{f'W\Delta f}{f'Wf} - \frac{f'\Delta f}{f'f}\right)^2,$$

which, after substituting the definitions (B.3), is exactly the expression in brackets in (B.2). □

G Robustness of the limiting distribution

These results extend Theorem 3.2 by relaxing its maintained assumptions; each is referenced from Section 3.1 of the main text.

Remark G.1 (Closed form for i.i.d. factors)**.** When $\Sigma_F = I_T$, $\gamma_{ab} = \gamma_{a+b}$ and (3.8) reduces to $\Omega = 2(\gamma_4 + 4\gamma_2^3 - \gamma_2^2 - 4\gamma_3\gamma_2)\rho_2^2 + 2(\gamma_2 - 1)^2(\rho_4 - \rho_2^2)$, and further to a closed form in σ^2 and c when e is also i.i.d.

Lemma G.2 (When the closed forms hold)**.** *Under **B2** with ε having i.i.d. entries, $\Delta = eM_\Lambda e'/N$ is a Wishart-type matrix whose eigenvectors are asymptotically Haar-distributed relative to any fixed deterministic $W = A'A$; hence Δ and W are asymptotically free (Anderson et al., 2009), and the factorization $\varphi(\Delta^k W^j) = \varphi(\Delta^k)\,\varphi(W^j)$ used to reduce the variance to (3.8) holds **for every deterministic** A—including the structured Toeplitz (KMS), autoregressive, and moving-average dips of the recommended library—provided the cross-sectional covariance Σ is itself free from W (in particular whenever $\Sigma = \sigma^2 I$, or more generally when Σ and W are not co-diagonalisable). Free-independence can fail only in the co-structured case where Σ shares eigenvectors with W (e.g. both Toeplitz), in which case (3.8) acquires additional mixed-moment terms.*

It is the **idiosyncratic** eigenbasis—Wishart, hence rotationally invariant in the limit—that supplies the free-independence, not the transformation; a structured A is therefore not an

obstacle to (3.8). And even in the co-structured case the feasible test is unaffected:

Proposition G.3 (The feasible variance requires no free-independence)**.** *The plug-in $\hat{\Omega}$ of Theorem 3.2 is formed by substituting $\hat{\Sigma}_F$, the Gram matrix $W = A'A$, and the eigenvalue moments $\hat{\rho}_k$ directly into the trace functionals $\hat{\tau}_k, \hat{\tau}_{ab}$ of the plug-ins above; it never invokes the factorization $\varphi(\Delta^k W^j) = \varphi(\Delta^k)\varphi(W^j)$. Consequently, if $\hat{\tau}_k - \tau_k = o_P(T^{-1/2})$ $(k = 1, 2)$ and $\hat{\tau}_{ab} - \tau_{ab} = o_P(1)$, then $\hat{\Omega} \xrightarrow{P} \Omega$ and the studentizedstatistic of Theorem 3.2 is asymptotically pivotal,* ***whether or not*** *Δ and W are asymptotically free. Free-independence serves only to produce the closed forms (3.8) and Remark G.1; the analyst estimates the mixed moments τ_{ab} from $\hat{\Sigma}_F$ and W as they are, structure and all.*

Proposition G.4 (Non-Gaussian factors and idiosyncratic terms)**.** *Relax* ***B1*** *to allow ξ with i.i.d. entries of mean 0, variance 1 and finite fourth moment, with excess kurtosis $\kappa_F = \mathbb{E}[\xi_1^4] - 3$ (the Gaussian case being $\kappa_F = 0$), and relax* ***B2*** *to i.i.d. mean-zero, unit-variance idiosyncratic entries with finite fourth moment. Then Theorem 3.2 continues to hold with Ω replaced by*

$$\Omega(\kappa_F) = \Omega + \kappa_F \, \Psi, \qquad \Psi = \mathrm{plim}_{N,T\to\infty} \frac{1}{T} \sum_{i=1}^{T} B_{s,ii}^2 \;\geq\; 0,$$

where Ω is the Gaussian variance (3.8) and $B_{s,ii}$ are the diagonal entries of the symmetric quadratic-form matrix B_s of Appendix B, in the time basis in which the factor innovations are i.i.d. The idiosyncratic distribution enters only through ρ_2, ρ_4 and leaves Ω unchanged. Both κ_F and Ψ are consistently estimable, the former by the sample excess kurtosis of the estimated factor innovations and the latter by $\hat{\Psi} = T^{-1} \sum_i \hat{B}_{s,ii}^2$.

The correction inflates the variance by $1 + \kappa_F \Psi/\Omega$, where Ψ is a delocalization functional that is small whenever B_s has spread-out entries, as it does for the smooth KMS dips: in our design $\Psi/\Omega \approx 0.007$, so even a strongly leptokurtic t_5 factor inflates the variance by under 4%. Together with the exact invariance of Ω to the idiosyncratic distribution, the limit law is robust to non-Gaussianity in both components; the $\kappa_F \Psi$ term matters only for strongly leptokurtic factors combined with a localizeddip (Online Appendix).

The i.i.d.-in-time form of **B2** is stronger than the theory needs, but temporal dependence in e interacts with the dip through exactly the free-independence condition of Lemma G.2, so the answer is conditional on the dip. Allow

B2′ (weak temporal dependence). $e_t = \Sigma^{1/2} \sum_{\ell \geq 0} \Psi_\ell \, \varepsilon_{t-\ell}$, where ε has i.i.d. mean-zero, unit-variance, finite-fourth-moment entries and $\sum_{\ell \geq 0} \|\Psi_\ell\| < \infty$ (short memory / α-mixing with summable coefficients). Write $\Sigma_e^{\mathrm{time}} = \mathrm{plim}\Delta$ for the resulting limiting idiosyncratic temporal covariance.

Proposition G.5 (Idiosyncratic serial dependence and free dips)**.** *Under **B1** and **B2′**: (a) if the dip is* asymptotically free from Σ_e^{time}*—in particular a random dip* $\tilde{A} = U_1 D_\sigma U_2'$*, or any dip when* e *is temporally i.i.d.* ($\Sigma_e^{\text{time}} \propto I$)*—then Theorem 3.2 holds verbatim, with* $\rho_k = \text{plim} T^{-1}\operatorname{tr}(\Delta^k)$ *now reflecting the dependence and estimated by the* same $\hat{\rho}_k = T^{-1}\sum_{j\geq 2}(\lambda_j/N)^k$*; (b) if instead the dip is* co-structured *with* Σ_e^{time}*—a temporal Toeplitz dip (KMS/AR/MA) applied to temporally persistent* e*, the freeness-failure case of Lemma G.2—the centring* ω_1 *no longer captures* $\text{plim}(\lambda_1/N)^2 S$*, the raw analytic test over-rejects, and the parametric-bootstrap standardization of the Online Appendix, resampling the estimated idiosyncratic dependence, re-centres the statistic and restores size.*

Proof sketch. Conditional on the idiosyncratic matrix (which enters only through Δ), the distributional statement is the quadratic-form CLT over the Gaussian factor draw (the Online Appendix), unaffected by dependence in e; the empirical spectral distribution of Δ and $\hat{\rho}_k$ remain consistent under $\sum_\ell \|\Psi_\ell\| < \infty$ (Bai and Silverstein, 1998). The reduction of the centring to ω_1 and of the variance to (3.8) uses $\varphi(\Delta^k W^j) = \varphi(\Delta^k)\varphi(W^j)$, which by Lemma G.2 holds when W is free from Σ_e^{time} and fails when they are co-diagonalisable; in the co-structured case the discrepancy is a pure centring shift that the bootstrap null of the Online Appendix reproduces and removes. □

The lesson is constructive: the *random* dips in the recommended library are free from any idiosyncratic temporal structure by construction and stay reliable under serially dependent e—one reason the library spans several families rather than the KMS one alone. The Online Appendix demonstrates both regimes numerically.

G.1 A log-transformed statistic

The level statistic $Q \equiv (\lambda_r/N)^2 S$ is non-negative and, in finite samples, right-skewed (the histograms of the Online Appendix make this visible). A variance-stabilising logarithmic transform improves the normal approximation while leaving the first-order theory and the power against weak factors unchanged. Under the null, Theorem 3.7 gives $\sqrt{T}\,(Q - \bar{\omega}) \xrightarrow{d} N(0, \bar{\Omega})$ with $\bar{\omega} = \sum_a (\lambda_r/\lambda_a)^2 \omega_1^{(a)} > 0$ and $\bar{\Omega} = \sum_a (\lambda_r/\lambda_a)^4 \Omega^{(a)}$. Since $x \mapsto \log x$ is continuously differentiable at $\bar{\omega} > 0$ with derivative $\bar{\omega}^{-1}$, the delta method gives

$$\sqrt{T}\left(\log Q - \log \bar{\omega}\right) \xrightarrow{d} N\Big(0,\ \bar{\Omega}/\bar{\omega}^2\Big), \qquad Z_{\text{blk}}^{\log} \equiv \frac{\sqrt{T}\,\hat{\bar{\omega}}\left(\log Q - \log \hat{\bar{\omega}}\right)}{\sqrt{\hat{\bar{\Omega}}}} \xrightarrow{d} N(0,1).$$

Because the transformation is strictly monotone, $Z_{\text{blk}}^{\log}$ delivers the same consistency and the same local power as Z_{blk}; its advantage is purely distributional. The log compresses the

heavy upper tail of Q that drives the finite-sample positive bias and right-skew of Z_{blk} (the Online Appendix), so $Z_{\text{blk}}^{\log}$ is typically more symmetric and better-sized in small samples—a cheap, fully analytic alternative to the simulation-based standardization. In Monte Carlo (Online Appendix) the log statistic is better sized and essentially symmetric, but once size is equalizedthe *level* statistic is more powerful (size-adjusted power 0.52 versus 0.43 against a single weak factor at $T = N = 200$). We retain the level statistic as the default and offer the log transform as an analytic option when symmetry of the reference distribution is at a premium.

H Proofs for the multi-factor extension

This appendix supplies the computations deferred from Appendix E of the main text, where the multi-factor limit theorems (Theorems 3.4, 3.6, 3.7, Lemma 3.5, Proposition 3.9) are proved: the group-projector block reduction, the moment-matrix limits and their single-index reduction (Lemma E.1), the exact reduction of $(\lambda_r/N)^2 S$ to the matrix bracket Ψ_Λ, and the identification counterexample. Notation is as in Appendix E: $g_a = \Sigma_{F,a}^{1/2}\xi_a$, $G = (g_1, \ldots, g_r)$, $R = (G'G/T)^{-1/2}$, $\bar{F} = GR$; kernels $V_1, \ldots, V_7$; $\widehat{\mathcal{Q}}_j = T^{-1}G'V_jG$, $\mathcal{Q}_j = R\widehat{\mathcal{Q}}_jR$. All limits are as $N, T \to \infty$, $T/N \to c$, conditionally on ε unless a probability limit over ε is indicated.

H.1 Block reduction (Theorem 3.4)

Write $\Phi = \Phi_0 + \Phi_1/n$, $\Phi_0 = \lambda_r^{-1}\mathcal{F}\mathcal{F}'$, with factor eigenprojectors $\pi_1, \ldots, \pi_r$ (eigenvalues $\mu_1 \geq \cdots \geq \mu_r$, possibly clustered) and idiosyncratic $\{\pi_k\}_{k>r}$. By **A1**, $\mu_r = \Theta_P(1)$ and the factor block is separated from the idiosyncratic spectrum by a $\Theta_P(1)$ gap; take a contour Γ enclosing exactly $\mu_1, \ldots, \mu_r$, so $P_0 = -\frac{1}{2\pi i}\oint_\Gamma (\Phi_0 - z)^{-1}dz = \sum_{i\leq r}\pi_i$ is the exact group projector. The Neumann series

$$(\Phi - z)^{-1} = \sum_{m\geq 0} n^{-m} R_0(z)[\Phi_1 R_0(z)]^m, \qquad R_0(z) = (\Phi_0 - z)^{-1} = \sum_k (\mu_k - z)^{-1}\pi_k,$$

converges uniformly on Γ because $n^{-1}\|\Phi_1\| \sup_{z\in\Gamma}\|R_0\| = o_P(1)$ (**A2**, $\mu_r = \Theta_P(1)$). Term-by-term integration gives $P_{\hat{F}} = P_0 + n^{-1}P^{(1)} + n^{-2}P^{(2)} + O_P(n^{-3})$, $P^{(m)} = -\frac{1}{2\pi i}\oint_\Gamma R_0[\Phi_1 R_0]^m dz$. By the residue theorem,

$$-\tfrac{1}{2\pi i}\oint_\Gamma (\mu_j - z)^{-1}(\mu_k - z)^{-1}dz = \begin{cases} 0, & j,k \leq r, \\ (\mu_j - \mu_k)^{-1}, & j \leq r < k, \end{cases}$$

the first case because both residues $(\mu_k-\mu_j)^{-1}$ and $(\mu_j-\mu_k)^{-1}$ lie inside Γ and cancel. Hence the within-block differences $(\mu_i-\mu_{i'})^{-1}$, $i,i'\le r$, drop out of $P^{(1)}$ (which reduces to block-to-noise couplings $\sum_{i\le r<k}(\mu_i-\mu_k)^{-1}[\pi_i\Phi_1\pi_k+\text{h.c.}]$) and, by the same cancellation of any two inside poles, out of $P^{(2)}$; the factor block enters only through P_0 and $\Phi_0^+=\sum_{i\le r}\mu_i^{-1}\pi_i$. The trace computation of Appendix B (the $r=1$ proof) therefore applies verbatim after summing over the block, and with the annihilation $M_0A^{-1}P_{A0}=0$ (valid because $A^{-1}\operatorname{range}P_{A0}=\operatorname{range}P_0$, which M_0 kills),

$$n^2S=\operatorname{tr}[P_0D_A\Phi_1M_0\Phi_1D_AP_0]+O_P(n^{-1})=\|P_0D_A\Phi_1M_0\|_F^2+O_P(n^{-1}),$$

$D_A=A'\Phi_{A0}^+A-\Phi_0^+$, $\Phi_0^+=\bar F\Lambda_{\text{blk}}\bar F'/T^2$. Only $n=\lambda_r/\lambda_{r+1}$ enters and the remainder is governed by $\mu_r=\Theta_P(1)$, so the reduction holds uniformly whether or not the factor eigenvalues are separated.

H.2 Moment-matrix limits and single-index reduction (Lemma E.1)

Diagonal limits. Fix ε, so Δ,W are deterministic with bounded norms. For $a\neq b$, $(\widehat{\mathcal{Q}}_j)_{ab}=T^{-1}\xi_a'\Sigma_{F,a}^{1/2}V_j\Sigma_{F,b}^{1/2}\xi_b$ is bilinear in independent mean-zero ξ_a,ξ_b, with $\mathbb{E}[\cdot\mid\varepsilon]=0$ and $\operatorname{Var}[\cdot\mid\varepsilon]=T^{-2}\operatorname{tr}(\Sigma_{F,a}^{1/2}V_j\Sigma_{F,b}V_j'\Sigma_{F,a}^{1/2})=O(T^{-1})$, so $(\widehat{\mathcal{Q}}_j)_{ab}=O_p(T^{-1/2})$. For $a=b$, the LLN for quadratic forms gives $(\widehat{\mathcal{Q}}_j)_{aa}\to_p T^{-1}\operatorname{tr}(\Sigma_{F,a}V_j)$, and the ε-limit follows from asymptotic freeness of the Wishart-type Δ from the fixed $\Sigma_{F,a},W$ (Anderson et al., 2009): e.g. $T^{-1}\operatorname{tr}(\Sigma_{F,a}W\Delta)\to\tau_1^{(a)}\rho_1$ and $T^{-1}\operatorname{tr}(\Sigma_{F,a}W\Delta^2W)=T^{-1}\operatorname{tr}(W\Sigma_{F,a}W\Delta^2)\to\tau_2^{(a)}\rho_2$, giving the $\bar q_j^{(a)}$ of Lemma E.1; since $R\to_p I_r$ the same holds for $\mathcal{Q}_j$.

Single-index reduction. Write $\frac{1}{T}G'G=I_r+\widehat E$, $\widehat E_{aa}=\delta\widehat q_1^{(a)}$ (a form in ξ_a), $\widehat E_{ab}=O_p(T^{-1/2})$. Then $R=\widehat{\mathcal{Q}}_1^{-1/2}=I_r-\frac12\widehat E+O_p(T^{-1})$, so $R_{aa}=1-\frac12\delta\widehat q_1^{(a)}+O_p(T^{-1})$ and $R_{ba}=-\frac12\widehat E_{ba}+O_p(T^{-1})$ $(b\neq a)$; off-diagonal Gram entries touch R_{aa} only at $O_p(T^{-1})$. With $\bar f_a=R_{aa}g_a+\sum_{b\neq a}R_{ba}g_b$,

$$(\mathcal{Q}_j)_{aa}=R_{aa}^2\tfrac1T g_a'V_jg_a+2R_{aa}\sum_{b\neq a}R_{ba}\tfrac1T g_a'V_jg_b+\sum_{b,c\neq a}R_{ba}R_{ca}\tfrac1T g_b'V_jg_c,$$

the last two sums being $O_p(T^{-1})$; with $R_{aa}^2=1-\delta\widehat q_1^{(a)}+O_p(T^{-1})$ this is $(\mathcal{Q}_j)_{aa}=\bar q_j^{(a)}+\delta\widehat q_j^{(a)}-\bar q_j^{(a)}\delta\widehat q_1^{(a)}+o_p(T^{-1/2})$, a quadratic form in ξ_a alone. This is the coupling control behind conditional independence: the orthonormaliser mixes directions only at $O_p(T^{-1})$ on the diagonal, negligible after $\sqrt{T}$ scaling.

H.3 Exact reduction to the matrix bracket

By Theorem 3.4, $\|P_0 D_A \Phi_1 M_0\|_F^2 = (N/\lambda_{r+1})^2 \operatorname{tr}[P_0 D_A \Delta M_0 \Delta D_A P_0]$. Substitute $P_0 = \frac{1}{T}\bar{F}\bar{F}'$, $M_0 = I - P_0$, $\Phi_0^+ = \bar{F}\Lambda_{\text{blk}}\bar{F}'/T^2$ and $A'\Phi_{A0}^+ A = \bar{F}_W \Lambda_{\text{blk}} \bar{F}_W'/T^2$, and contract on $\bar{F}$; every factor $\bar{F}'V\bar{F}/T$ is one of the $\mathcal{Q}_j$. Carrying the one-factor algebra of Appendix B on these $r \times r$ blocks — the only change from $r = 1$ being that the squared term is a genuine Frobenius square $\|M\|_F^2$, non-commuting for $r > 1$ — yields $(\lambda_r/N)^2 S = \Psi_\Lambda(\mathcal{Q}) + O_p(n^{-1})$ with $D = \Lambda_{\text{blk}}/\lambda_r$. The strength matrix D enters only through $\Phi_0^+ \propto \Lambda_{\text{blk}}$, i.e. as a fixed diagonal weight, and the diagonal restriction of Ψ_Λ is $\sum_a (\lambda_a/\lambda_r)^2 g(1, q_2^{(a)}, \ldots, q_7^{(a)})$, the sum of one-factor brackets.

Remark H.1 (Numerical corroboration of Lemma E.2)**.** The off-diagonal annihilation of Lemma E.2 was verified symbolically: at every diagonal $\bar{M} = 0$ point, and for randomised r and moment values, the partial derivatives of Ψ_Λ with respect to the off-diagonal entries of each $\mathcal{Q}_j$ vanish identically, while the diagonal partials are generically nonzero.

H.4 Off-diagonal annihilation, linearisation, and the joint limit (Lemma E.2; Theorems 3.6–3.7)

Proof of Lemma E.2 (off-diagonal annihilation). $\bar{M} = 0$ since $\bar{\mathcal{Q}}_2^{-1}\bar{\mathcal{Q}}_4 = \operatorname{diag}(\tau_1^{(a)})^{-1} \operatorname{diag}(\tau_1^{(a)}\rho_1) = \rho_1 I_r = \bar{\mathcal{Q}}_3$. Because $\bar{M} = 0$, the differential of $-\operatorname{tr}[MM']$ is $-2\operatorname{tr}[\bar{M}\,\delta M'] = 0$, so $\mathcal{Q}_3, \mathcal{Q}_4$ drop out ($w_3^{(a)} = w_4^{(a)} = 0$). In every remaining term each non-perturbed factor is one of the diagonal matrices $D, \bar{\mathcal{Q}}_2^{-1}, \bar{\mathcal{Q}}_5, \bar{\mathcal{Q}}_6, \bar{\mathcal{Q}}_7$; using $\delta(\mathcal{Q}_2^{-1}) = -\bar{\mathcal{Q}}_2^{-1}\delta\mathcal{Q}_2\bar{\mathcal{Q}}_2^{-1}$, each first-order piece is $\operatorname{tr}[\Xi\,\delta\mathcal{Q}_j] = \sum_a \Xi_{aa}(\delta\mathcal{Q}_j)_{aa}$ for a diagonal Ξ, so off-diagonal entries do not enter. Restricting Ψ_Λ to diagonal arguments gives $\sum_a (\lambda_a/\lambda_r)^2\, g(1, q_2^{(a)}, \ldots, q_7^{(a)})$, identifying the one-factor weights $w_j^{(a)} = (\lambda_a/\lambda_r)^2\, \partial g/\partial q_j$.

Linearisation. A second-order Taylor expansion of Ψ_Λ about the diagonal limit has quadratic term $O_p(\|\delta\mathcal{Q}\|^2) = O_p(T^{-1})$—exactly the cross-direction remainder $\mathcal{E}_T$ of (3.10), so $\sqrt{T}(\lambda_r/N)^2\mathcal{E}_T = o_p(1)$ (Lemma 3.5(ii)). By Lemma E.2 the linear term keeps only the diagonal fluctuations, and by Lemma E.1 each $(\mathcal{Q}_j)_{aa} - \bar{q}_j^{(a)} = \delta\widehat{q}_j^{(a)} - \bar{q}_j^{(a)}\delta\widehat{q}_1^{(a)} + o_p(T^{-1/2})$ is a quadratic form in ξ_a alone. Substituting the weights $w_j^{(a)}$ yields

$$\sqrt{T}\Big[\big(\tfrac{\lambda_r}{N}\big)^2 S - \textstyle\sum_a \big(\tfrac{\lambda_r}{\lambda_a}\big)^2 \omega_1^{(a)}\Big] = \textstyle\sum_{a=1}^r \big(\tfrac{\lambda_r}{\lambda_a}\big)^2 U_a + o_p(1),$$

where $U_a = \sqrt{T}[(\lambda_a/N)^2 S_a - \omega_1^{(a)}]$ is verbatim the one-factor delta-method statistic of Appendix B at $\Sigma_{F,a}$ (the term $-\bar{q}_j^{(a)}\delta\widehat{q}_1^{(a)}$ supplying the one-factor $Q_1 = f'f$ contribution). This is Lemma 3.5(i)–(ii).

Joint limit and independence (Lemma 3.5(iii); Theorems 3.6–3.7). Condition on ε: $U_a = \sqrt{T}(\xi_a' B_T^{(a)} \xi_a / T - T^{-1} \operatorname{tr} B_T^{(a)}) + o_p(1)$ with $B_T^{(a)}$ the one-factor matrix of Appendix B at $\Sigma_{F,a}$. *(i)* Given ε, each U_a is a measurable function of ξ_a only, and the ξ_a are independent (**B1′**); hence $U_1, \ldots, U_r$ are conditionally independent—derived, not assumed. *(ii)* The quadratic-form CLT of Appendix B holds conditionally, $U_a \mid \varepsilon \xrightarrow{d} N(0, \Omega_\varepsilon^{(a)})$, and $\Omega_\varepsilon^{(a)} \xrightarrow{p} \Omega^{(a)}$ (its ε-dependence is only through ρ_2, ρ_4, which concentrate). With $\phi_a(t;\varepsilon) = \mathbb{E}[e^{itU_a} \mid \varepsilon]$, for any $t \in \mathbb{R}^r$,

$$\mathbb{E}\big[e^{i\sum_a t_a U_a}\big] = \mathbb{E}\big[\textstyle\prod_a \phi_a(t_a;\varepsilon)\big] \to \prod_a e^{-t_a^2 \Omega^{(a)}/2}$$

by bounded convergence ($|\phi_a| \le 1$, each $\to_p$ a constant), so $(U_1, \ldots, U_r) \xrightarrow{d} N(0, \operatorname{diag}(\Omega^{(a)}))$: joint normality *and* asymptotic independence. The weighted sum gives Theorem 3.7; Case H (Theorem 3.6) is $\Sigma_{F,a} \equiv \Sigma_F$, collapsing to $N(0, r\Omega)$ at $r\omega_1$ when the λ_a are equal, and $Z_{\text{blk}} \xrightarrow{d} N(0,1)$ by Slutsky.

H.5 Identification boundary: a counterexample

By (3.8), $\Omega^{(a)}$ is affine in $\operatorname{tr}(\Sigma_{F,a} W^c \Sigma_{F,a} W^d)$, so $\sum_a \Omega^{(a)}$ depends on $\sum_a \operatorname{tr}(\Sigma_{F,a} W \Sigma_{F,a} W)$, not on $\bar{\Sigma} = \sum_a \Sigma_{F,a}$ alone. Take $r = 2$ and W with two distinct eigenvalues: $\{\Sigma_{F,1}, \Sigma_{F,2}\}$ and $\{\frac{1}{2}(\Sigma_{F,1}+\Sigma_{F,2}), \frac{1}{2}(\Sigma_{F,1}+\Sigma_{F,2})\}$ share $\bar{\Sigma}$ but give $\sum_a \operatorname{tr}(\Sigma_{F,a} W \Sigma_{F,a} W)$ differing by $\frac{1}{2}\operatorname{tr}((\Sigma_{F,1} - \Sigma_{F,2}) W (\Sigma_{F,1} - \Sigma_{F,2}) W) \neq 0$ whenever $\Sigma_{F,1} - \Sigma_{F,2}$ does not commute with W. Under clustered eigenvalues only $\bar{\Sigma}$-measurable quantities are estimable, so $\sum_a \Omega^{(a)}$ is unidentified and only Case H is feasible; the separated regime identifies each $\Sigma_{F,a}$ and Case K applies. This is Proposition 3.9.

I Simulation design and additional Monte Carlo evidence

I.1 Size and power of the specification test

We evaluate the multi-factor test and the dipping-based uncertainty assessment by Monte Carlo. The panel is $X = F\Lambda' + e$ ($T \times N$), with r factors of heterogeneous strength (the k-th factor eigenvalue of XX' grows as N^{α_k}, $\alpha_k \in (0,1]$, so $\alpha_k = 1$ is strong and $\alpha_k < 1$ weak), stationary ARMA temporal dependence that is either common across factors (Case H) or factor-specific (Case K), idiosyncratic term $e = \varepsilon \Sigma^{1/2}$ with $\Sigma_{ij} = \rho_e^{|i-j|}$, and the representative Kac–Murdock–Szegő dip $a = 0.3$ ($\|A\| \approx \|A^{-1}\| \approx 1.9$). The full designs, the moments of Z_{blk} and its QQ plots, the near-unit-root boundary study, and the additional experiments

summarizedbelow are collected in the Online Appendix; here we report the headline size and power.

Size. Table 5 places the *raw* analytic test (centred and scaled by the plug-in $\hat{\omega}_1, \hat{\Omega}$) beside the *feasible* simulation-standardizedtest, which re-centres and re-scales each transformation's summand by its null mean and standard deviation simulated from the maintained strong-factor model. The raw test is near nominal for i.i.d. factors but over-rejects under temporal dependence and factor heterogeneity, worst in small samples (Case K, 0.31 at $(100, 100)$); the residual distortion is a positive finite-sample *centring* bias—the standardizedstatistic keeps standard deviation near one throughout—which shrinks with T. The standardizedtest is at nominal across *every* design and sample size (0.036–0.063), and leaves power essentially intact. This is the test we recommend when many factors are entertained in a small, temporally dependent panel. Cross-sectional dependence in e ($\rho_e \in \{0, 0.3, 0.5\}$) barely moves the size, confirming that it enters only through the idiosyncratic moments ρ_2, ρ_4.

Table 5: Consolidated size: raw analytic test versus the feasible simulation-standardizedtest ($r = 3$ strong factors, $\alpha = (1, 1, 1)$, dip $a = 0.3$, nominal 5%, two-sided; no cross-sectional dependence). Each cell is a fresh Monte Carlo run (600/500/250 replications at $T = 100/200/500$, with an equal-size independent maintained-null run for the standardization); the "raw" columns reproduce the Online Appendix. The standardizedtest is at nominal in every design and at every sample size.

	$(100, 100)$		$(200, 200)$		$(500, 500)$	
Σ_F structure	raw	standardized	raw	standardized	raw	standardized
i.i.d. factors	0.065	0.040	0.060	0.052	0.072	0.036
ARMA(1,1)	0.252	0.057	0.204	0.046	0.104	0.044
ARMA(1,2)	0.303	0.045	0.214	0.052	0.152	0.036
Heterogeneous (Case K)	0.310	0.063	0.270	0.038	0.216	0.056

Power. Power against weak-factor misspecification at $T = 200$ (Table 12) rises sharply as factors weaken. Weakening factors raises rejection sharply: a single half-strength factor lifts power to about 0.5, two weak factors to 0.7–0.8, and a markedly weak factor to essentially one; power grows with N, as a wider cross-section sharpens the separation the dip probes. The power is a bodily rightward migration of the null distribution rather than a knife-edge, which is why it rises so steeply (Online Appendix).

The spectral gap governs reliability. Size, power, and the near-unit-root boundary are all manifestations of the gap $n = \lambda_r/\lambda_{r+1}$ relative to $\sqrt{T}$: at a borderline strength $\alpha = 0.5$

the realized$n/\sqrt{T} \approx 0.2$ determines whether the Gaussian approximation has taken hold, so a larger T sharpens size and power while a near-unit-root factor, which shrinks the gap, degrades the test at any sample size (size rises from 0.13 at $\phi = 0.5$ to 1 at $\phi = 0.99$; Online Appendix). The finite-sample bias is $O_p(n^{-1})$, largest when the gap is narrow, and has a transformation-side counterpart: a near-differencing dip inflates $\|A^{-1}\|$ and weakens the transformed factor through the same channel.

This appendix gives the finite-sample design behind Section 4 of the main text and the additional experiments summarizedthere.

I.2 Non-Gaussian factors and idiosyncratic terms

Assumptions **B1**–**B2** impose Gaussian factors and idiosyncratic entries for the distributional theory, but Proposition G.4 shows the limit is robust to both: the idiosyncratic distribution enters only through ρ_2, ρ_4 and leaves Ω unchanged, while factor kurtosis contributes only the term $\kappa_F \Psi$ with Ψ a delocalization functional that is small for smooth dips. We verify this directly. Table 6 reports the size of the one-factor test ($r = 1$, strong factor, $\Sigma_F = \Sigma = I$, representative dip $a = 0.3$, $T = N = 200$, 1500 replications) under Gaussian and standardised-t_5 draws (excess kurtosis $\kappa_F = 6$) for the factor and for the idiosyncratic component separately, using the Gaussian variance (3.8) and, in the last row, the kurtosis-corrected $\Omega(\kappa_F) = \Omega + \kappa_F \Psi$.

Table 6: Empirical size under non-Gaussian factors and idiosyncratic terms ($r = 1$, strong factor, $\Sigma_F = \Sigma = I$, dip $a = 0.3$, $T = N = 200$, 1500 replications). Standardised-t_5 draws have excess kurtosis $\kappa_F = 6$. All cells carry the same positive finite-sample centring bias as the Gaussian baseline; neither non-Gaussian factors nor non-Gaussian idiosyncratic terms move the size beyond it, and the kurtosis correction $\Omega + \kappa_F \Psi$ is negligible ($\Psi/\Omega \approx 0.007$).

Design	variance used	size (5%)	size (10%)
factor N, idiosyncratic N	Ω (Gaussian)	0.089	0.141
factor N, idiosyncratic t_5	Ω (Gaussian)	0.089	0.140
factor t_5, idiosyncratic N	Ω (Gaussian)	0.083	0.132
factor t_5, idiosyncratic N	$\Omega + \kappa_F \Psi$ (corrected)	0.082	0.137

Three conclusions follow. First, replacing Gaussian idiosyncratic entries by t_5 (row 2) leaves the size identical to the Gaussian baseline (row 1), confirming that the idiosyncratic distribution does not enter Ω; **B2** may be dropped to a finite-fourth-moment condition. Second, a heavy-tailed t_5 *factor* (row 3) also leaves the size essentially unchanged, because for the smooth Kac–Murdock–Szegő dip the delocalization functional is small ($\Psi/\Omega \approx 0.007$), so the variance inflation $1 + \kappa_F \Psi/\Omega$ is under 4% even at $\kappa_F = 6$. Third, applying the correc-

tion (row 4) changes nothing, as expected. The residual over-size common to all four rows (≈ 0.089 at nominal 5%) is the pre-existing Gaussian finite-sample centring bias documented in the main text, not a distributional effect, and is removed by the simulation-based standardization of the main text. The Gaussian limit law is therefore an accurate reference under substantial non-Gaussianity in either component; the κ_F term is available (Proposition G.4) for the atypical combination of a strongly leptokurtic factor with a localised, near-diagonal transformation.

Serially dependent idiosyncratic terms. Proposition G.5 predicts that temporal dependence in e is benign for a dip that is free from it but not for a co-structured one. A one-factor experiment ($r = 1$, strong factor, $\Sigma_F = I$, $T = N = 200$, cross-sectionally i.i.d. but temporally AR(1) idiosyncratic $e_t = \phi_e e_{t-1} + u_t$) confirms both halves sharply. With the temporal KMS dip $a = 0.3$—co-structured with the AR(1) temporal covariance, both being Toeplitz—the raw analytic test is badly over-sized once $\phi_e > 0$: nominal-5% rejection rises from 0.08 at $\phi_e = 0$ to 0.76 at $\phi_e = 0.3$, the failure being a centring shift (mean of Z climbs from ≈ 0.1 to ≈ 3.0). A *random* dip $\tilde{A} = U_1 D_\sigma U_2'$ applied to the *same* data is essentially unaffected—rejection 0.09 at $\phi_e = 0.3$—because it is asymptotically free from the idiosyncratic temporal structure. The two dips see the identical panel; only their relationship to Σ_e^{time} differs, exactly as Lemma G.2 and Proposition G.5 require. In practice this means (i) the random-dip family of the recommended library is the safe choice under suspected idiosyncratic serial dependence, and (ii) when a temporal dip is nonetheless used, the bootstrap standardization of the main text—which resamples the estimated dependence—restores nominal size.

I.3 Near-differencing dips and spurious factors

The near-unit-root boundary above has a transformation-side counterpart. Here we examine how factor dipping behaves on integrated data, and turn the near-differencing dip into a probe for a phenomenon that ordinary dips cannot see: the *spurious* factors that principal components manufacture from factorless nonstationary panels (Onatski and Wang, 2021).

Boundedness is relative to the diverging dimension. Assumption A3 requires $\|A\|$ and $\|A^{-1}\|$ to be $O_P(1)$ as $T \to \infty$; "bounded" here means bounded *relative to the diverging sample dimension*, not by any fixed constant. The quasi-differencing family $A(\rho) = I - \rho L$ makes this concrete: its inverse is the lower-triangular operator $(A(\rho)^{-1})_{ij} = \rho^{i-j}$ for $i \geq j$, with $\|A(\rho)\| \leq 1 + \rho$ and $\|A(\rho)^{-1}\| \to (1-\rho)^{-1}$ as $T \to \infty$. For any fixed $\rho < 1$ both norms are constants in T—the transform satisfies A3—whereas exact first-differencing ($\rho = 1$) has

$\|A^{-1}\|$ of order T and violates it. Table 7 confirms this directly: for $\rho < 1$ the inverse norm stabilises as T grows (e.g. $9.8 \to 10.0$ at $\rho = 0.9$), while at $\rho = 1$ it doubles with the sample ($96 \to 191 \to 382$ for $T = 150, 300, 600$). We may therefore take ρ as close to differencing as we wish and remain inside A3, with the bound $(1-\rho)^{-1}$ growing but finite.

Table 7: Spectral norms of the near-differencing transform $A(\rho) = I - \rho L$. For every $\rho < 1$ the inverse norm $\|A(\rho)^{-1}\|$ is bounded in T, converging to $(1-\rho)^{-1}$; only exact differencing $\rho = 1$ gives $\|A^{-1}\|$ of order T (it doubles as T doubles). The direct norm is $\|A(\rho)\| \approx 1 + \rho$ throughout.

	$\|A(\rho)^{-1}\|$				
ρ	$T = 150$	$T = 300$	$T = 600$	$(1-\rho)^{-1}$	$\|A(\rho)\|$
0.90	9.8	9.9	10.0	10	1.90
0.99	57.0	77.6	91.2	100	1.99
0.999	90.3	170.1	304.4	1000	2.00
1.000 (differencing)	95.8	191.3	382.3	∞	2.00

Near-differencing transforms and the A3 boundary. Assumption A3 excludes exact first-differencing because its inverse, cumulation, has operator norm of order T. We probe this boundary directly with the quasi-differencing family $A(\rho) = I - \rho L$ (L the lag operator), which tends to first-differencing as $\rho \to 1$ while $\|A(\rho)^{-1}\| \approx (1-\rho)^{-1} \to \infty$. Figure 4 traces the resulting subspace deviation S for integrated (I(1), random-walk) and stationary factors of varying strength. For *strong* factors—including strongly-trending I(1) factors—the dipped estimate preserves the PC factor space essentially perfectly ($S \approx 0$, minimum canonical correlation ≈ 1) all the way to exact differencing, even though A3 is violated and $\|A^{-1}\|$ is large. This is precisely the logic of the PANIC procedure of Bai and Ng (2004): differencing an integrated panel, extracting factors, and cumulating back recovers the same space. The bounded-inverse condition is therefore *sufficient but not necessary*—a worst-case guard that binds only when the factor is weak *after* the transformation. The relevant strength is that of the transformed factor: degradation requires the common component to be weak *after* differencing—no longer dominant over the (also differenced) idiosyncratic noise. A weak *stationary* factor degrades in this way, its minimum canonical correlation falling toward 0.5–0.76 as $\rho \to 1$. A merely small *loading* on an I(1) factor does not suffice, however: integration keeps such a factor strong even after differencing, so it remains protected (Table 8). In the cases that do degrade, the deviation grows monotonically with ρ, tracking $\|A^{-1}\|$—the idiosyncratic amplification A3 is designed to prevent. The constructive reading is that for integrated data with strong factors, differencing-type dips are not merely admissible in spirit

but empirically harmless, exactly as PANIC presumes.

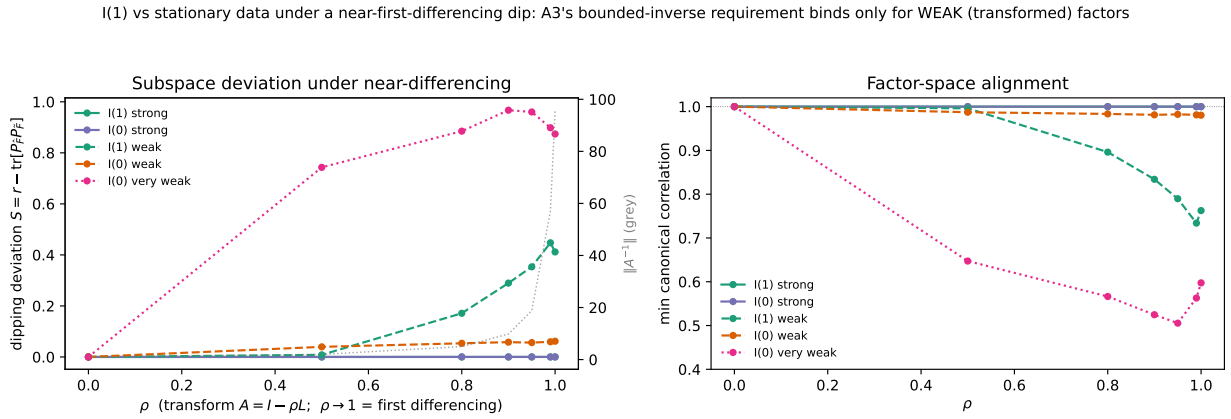


Figure 4: Near-differencing dip $A(\rho) = I - \rho L$ ($\rho \to 1$ is first-differencing) on integrated and stationary panels ($T = 150$, $N = 100$, $r = 3$). Left: subspace deviation $S = r - \mathrm{tr}[P_{\hat{F}} P_{\tilde{F}}]$ (grey dotted: $\|A^{-1}\|$, right axis). Right: minimum canonical correlation between the levels-PC and dipped estimates. Strong factors—including I(1)—are preserved up to exact differencing; only weak transformed factors degrade.

Spurious factors from nonstationarity. A sharper use of the near-differencing dip emerges when there are *no* factors at all. Onatski and Wang (2021) show that principal components extracted from a panel of independent nonstationary series manufacture *spurious* factors: a few components absorb almost all the variation, the scree plot and the Bai and Ng (2002) information criteria indicate several factors, and unit-root tests on the estimated idiosyncratic terms reject—in data with no common structure whatsoever. We reproduce this in a panel of N independent random walks: the leading three eigenvalues carry 93% of the variation, the eigenvalue-ratio selector returns one factor, and $\mathrm{IC}_{p1}, \mathrm{IC}_{p2}$ select the maximum number permitted, so the scree is indistinguishable from a genuine factor model (Figure 5, panel A).

Factor dipping interacts with these spurious factors in a way that depends entirely on the dip. Under *ordinary* dips—the bounded KMS library $\mathbf{A}_T(a)$ used throughout this paper—the spurious factor space is preserved exactly: the subspace deviation is $S \approx 0$ and the minimum canonical correlation is ≈ 1, indistinguishable from a genuine strong factor (Table 8). This is itself the Onatski–Wang warning in our language: a spurious factor is, in finite samples, a strong eigen-direction, and a bounded boundedly-invertible transform preserves strong eigen-directions whether or not they correspond to real common structure. The ordinary specification test, like the scree plot and the information criteria, cannot see the problem.

A *near-differencing* dip can. Because the spurious factor lives only in the levels—it is

an artefact of integration—differencing returns independent increments with no common component, so the dipped space detaches: S rises to about 1.0 and the minimum canonical correlation collapses to ≈ 0.4 as $\rho \to 1$, stably across $N \in \{100, \ldots, 500\}$ (Figure 5, panels B–C). Crucially, this is *not* merely the idiosyncratic amplification of A3: a genuine I(1) common factor, and even a weak *stationary* factor, are preserved by the same near-differencing dip ($S \approx 0$, canonical correlation ≥ 0.96 in Table 8), because a genuine common factor—unlike a spurious one—remains common after differencing. The near-differencing dip thus separates the case that defeats the standard toolkit: structure that is common at all frequencies (genuine) from structure that is common only through a shared unit root (spurious).

Table 8: Minimum canonical correlation between the levels-PC space and the dipped space (mean over replications; $T = 150$, $N = 100$, $r = 3$; ordinary KMS dip $a = 0.6$, near-differencing dip $\rho = 0.99$). Ordinary dips preserve both genuine and spurious factors; near-differencing preserves genuine factors—strong or weak, I(0) or I(1)—but detaches spurious ones.

data-generating process	spectral gap n	ordinary dip ($a = 0.6$)	near-diff dip ($\rho = 0.99$)
genuine I(1) strong factor	190	1.00	1.00
genuine I(0) strong factor	24	1.00	1.00
genuine I(0) weak factor ($\alpha = 0.5$)	3.6	0.96	1.00
spurious (independent random walks)	2.0	1.00	0.41
i.i.d. (no factor)	1.0	0.15	0.36

What this does and does not deliver. We do not claim a formal test against spurious factors; we report honestly what dipping does. Two messages follow. First, ordinary dips inherit the blind spot of PCA on nonstationary data—a useful caution, since it says the specification test should be read alongside a unit-root pretest when the data may be integrated. Second, the two families of dip are *complementary* diagnostics: ordinary dips detect weak factors but not spurious ones, while a near-differencing dip detects spurious factors but preserves genuine ones, weak or strong, stationary or integrated. A factor space that survives the KMS library yet collapses under a near-differencing dip is the signature of the Onatski–Wang phenomenon. This operationalises, within the dipping framework, the message of Onatski and Wang (2021) together with the PANIC logic of Bai and Ng (2004): differencing is the natural lens for integrated data, and dipping makes the consequences of not using it visible. The caveats are equally honest: the spurious factors also carry a smaller eigenvalue gap ($n \approx 2$ versus ≈ 190 for the genuine factor), which is itself a hint; cross-correlation in the increments would blur the separation; and a panel with no factors at all (i.i.d.) inflates S under every dip, a distinct and trivial signal.

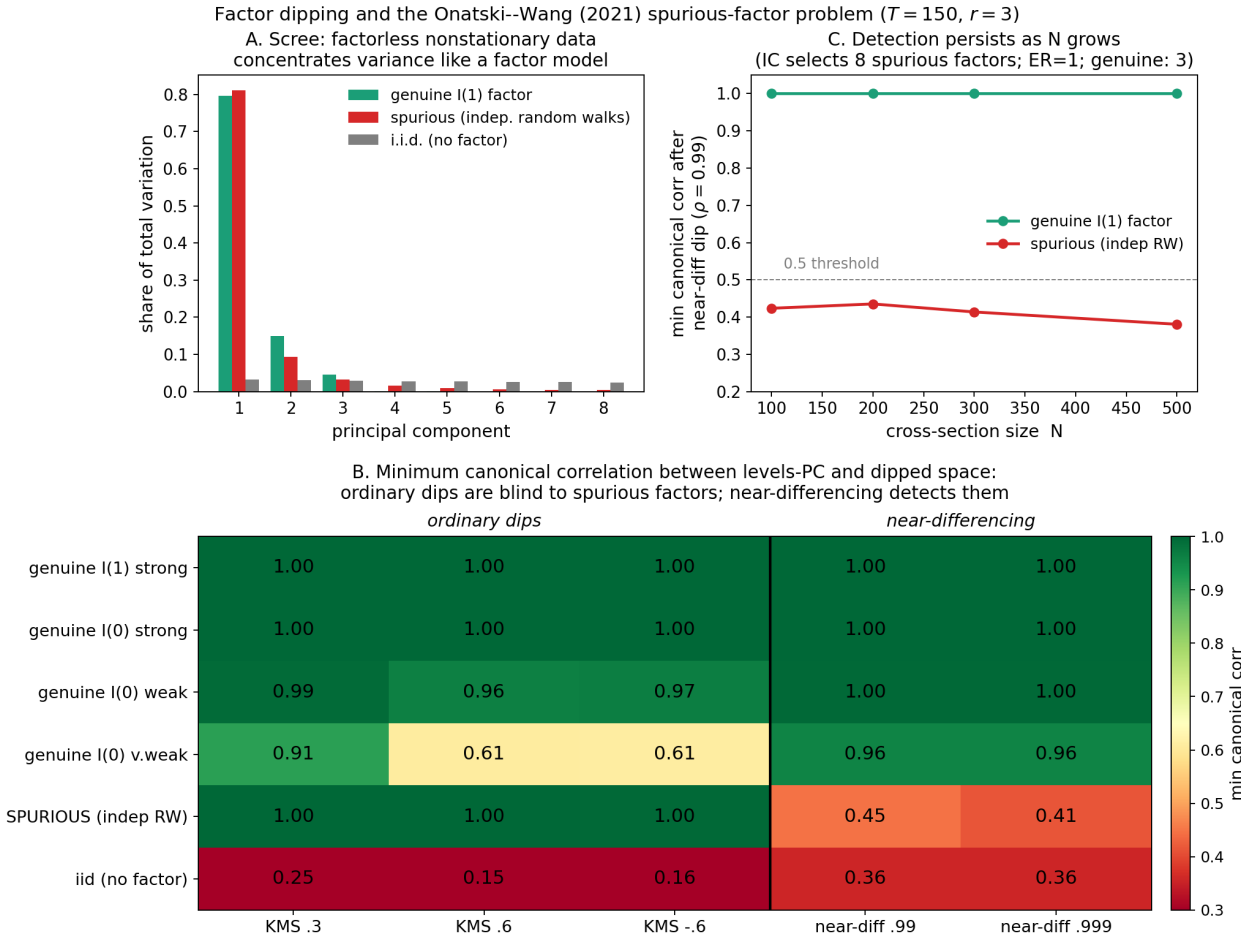


Figure 5: Factor dipping and the spurious-factor problem of Onatski and Wang (2021) ($T = 150$, $r = 3$). **A:** scree—a panel of independent random walks (no factor) concentrates variance much like a genuine factor model. **B:** minimum canonical correlation by data-generating process and transform; ordinary KMS dips preserve spurious factors (blind to them), whereas near-differencing dips detach them while preserving genuine factors, including weak ones. **C:** the near-differencing separation persists as N grows—the regime in which the information criteria select the spurious factors.

Table 9: Finite-sample reliability of $\hat{r}_{\text{dip}}$ on all-strong designs ($|a| \leq 0.6$ library, 20 replications): the fraction of replications that recover the true r (the modal $\hat{r}_{\text{dip}}$ equals the true r in every cell). In the small panel the uncorrected reliability declines as r grows; the finite-sample standardization restores it, and the large panel is reliable without correction.

true r	small panel, uncorrected $(T, N) = (150, 100)$	small panel, standardised $(150, 100)$	large panel $(500, 350)$
2	0.95	1.00	0.95
3	0.90	0.95	1.00
4	0.90	0.95	1.00
6	0.75	0.95	0.95
8	0.70	0.90	1.00

I.4 Finite-sample design

This appendix collects the finite-sample evidence summarised in the main text: the growth of the uncorrected over-rejection with the number of factors r and the weakness of the factors, and the restoration of nominal behaviour by the finite-sample standardization.

Table 10 reports the rejection rate of the (uncorrected, oracle-variance) omnibus test as a function of the common factor strength α and the number of factors r, at $T = N = 200$. At $\alpha = 1$ the size already departs from nominal as r grows—0.08 at $r = 1$ but 0.66 at $r = 8$—and deteriorates further as α falls and the spectral gap $n = \lambda_r/\lambda_{r+1}$ narrows. The breakdown tracks $n/\sqrt{T}$: reliability requires $n \gg \sqrt{T}$ (Assumption B4), and at $T = 200$ this holds comfortably only for strong, few factors. This is the bias that the standardization removes and that vanishes as $N, T \to \infty$.

Table 10: Rejection rate of the uncorrected omnibus test under correct specification with r equally strong factors of common strength α ($T = N = 200$, dip $a = 0.5$, nominal 5%, 500 replications). Mean spectral gap n in parentheses for $\alpha \in \{1.0, 0.6, 0.3\}$. The over-rejection grows with r and as α falls.

α	1.0	0.9	0.8	0.7	0.6	0.5	0.4	0.3
$r = 1$	0.08	0.09	0.10	0.13	0.20	0.35	0.62	0.89
$r = 3$	0.19	0.23	0.28	0.38	0.55	0.78	0.97	1.00
$r = 5$	0.32	0.38	0.46	0.57	0.79	0.95	1.00	1.00
$r = 8$	0.66	0.72	0.79	0.91	0.98	1.00	1.00	1.00
mean gap n ($r = 3$)	36.4	21.6	12.9	7.8	4.8	3.0	2.0	1.4

Table 11 gives the modal $\hat{r}_{\text{dip}}$, its mean, and the fraction of replications recovering the true r ("% correct"), for the all-strong designs across the three regimes. The small-panel

uncorrected test collapses for $r \geq 4$; the standardization restores the true count with 95–100% recovery, and the large panel is accurate without correction.

Table 11: Recovery of the true factor number by $\hat{r}_{\text{dip}}$ on all-strong designs, across the three regimes. Entries are the modal count, the mean (in parentheses), and the fraction of replications equal to the true r.

	small, uncorrected $(150, 100)$			small, standardized$(150, 100)$			large $(500, 350)$		
true r	mode	mean	%corr	mode	mean	%corr	mode	mean	%corr
2	2	1.70	0.85	2	1.80	0.90	2	1.80	0.90
3	3	2.55	0.85	3	3.00	1.00	3	3.00	1.00
4	0	2.00	0.50	4	3.80	0.95	4	3.40	0.85
6	0	2.65	0.40	6	6.00	1.00	6	5.10	0.85
8	0	2.00	0.25	8	7.60	0.95	8	6.80	0.85

Finally, the standardization preserves the ability to drop genuinely weak factors: in the small standardizedpanel the mixed designs with one weak factor ($\alpha_w = 0.4$) yield $\hat{r}_{\text{dip}} = r - 1$ (three-strong-plus-one-weak gives 3, five-strong-plus-one-weak gives 5, seven-strong-plus-one-weak gives 7), exactly as in the large panel—the correction removes the size distortion without sacrificing power.

Table 12: Power against weak-factor misspecification ($T = 200$, AR(1) factors, cross-sectional dependence $\rho_e = 0.3$, dip $a = 0.3$, nominal 5%). First row: correctly specified all-strong design (size). Power rises as factors weaken and grows with N.

strength α	$N = 100$	$N = 200$	$N = 400$	$N = 800$
$(1, 1, 1)$ (size)	0.132	0.116	0.138	0.158
$(1, 1, 1/2)$	0.530	0.480	0.532	0.554
$(1, 1/2, 1/2)$	0.758	0.770	0.706	0.808
$(1, 1/2, 1/4)$	0.990	0.998	1.000	1.000

I.5 Additional figures

This appendix collects the two distributional diagnostics not shown in the main text (the size-by-structure, bias-with-r, power, and standardization histograms appear in Section 4). Each histogram is the Monte Carlo distribution of the omnibus statistic Z_{blk}, overlaid with the $N(0, 1)$ reference.

Figure 8 shows that under correct specification with strong i.i.d. factors the statistic tightens onto $N(0, 1)$ as the sample grows, the small positive mean (the finite-sample bias)

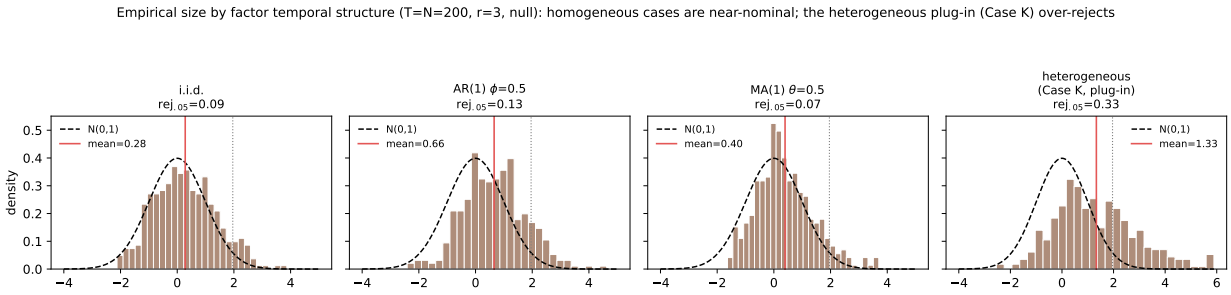


Figure 6: Null distribution of Z_{blk} by factor temporal structure ($T = N = 200$, $r = 3$, dip $a = 0.3$), with the $N(0,1)$ reference (dashed) and the mean (red). The i.i.d. design is near-nominal; ARMA(1,1), ARMA(1,2), and the heterogeneous (oracle Case K) design carry a progressively larger positive finite-sample location bias that the standardization removes.

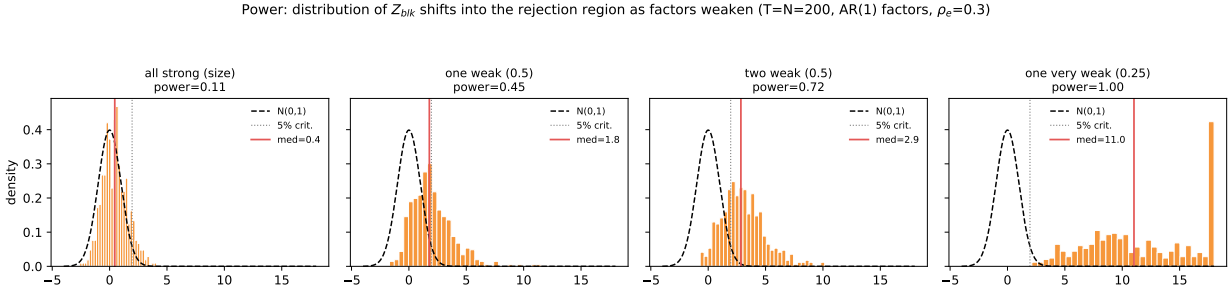


Figure 7: Distribution of Z_{blk} as factors weaken ($T = N = 200$, AR(1) factors, $\rho_e = 0.3$). Left to right: all strong (size), one weak, two weak, one very weak. The distribution migrates bodily past the 5% critical value (dotted); the rejection probability rises from ≈ 0.12 to near one.

collapsing toward zero. Figure 9 shows the statistic inflating as the AR(1) factor persistence $\phi \to 1$, the distributional counterpart of the near-unit-root size table.

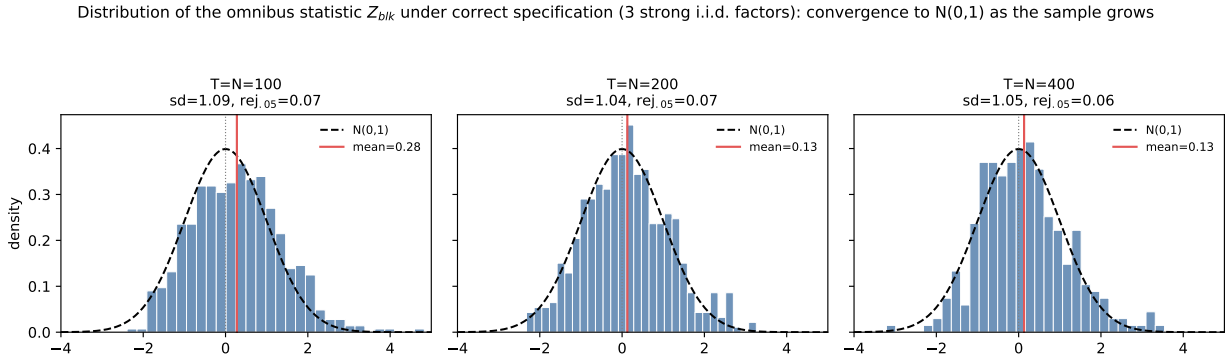


Figure 8: Convergence of Z_{blk} to $N(0,1)$ under correct specification ($r = 3$ strong i.i.d. factors) as $T = N$ grows.

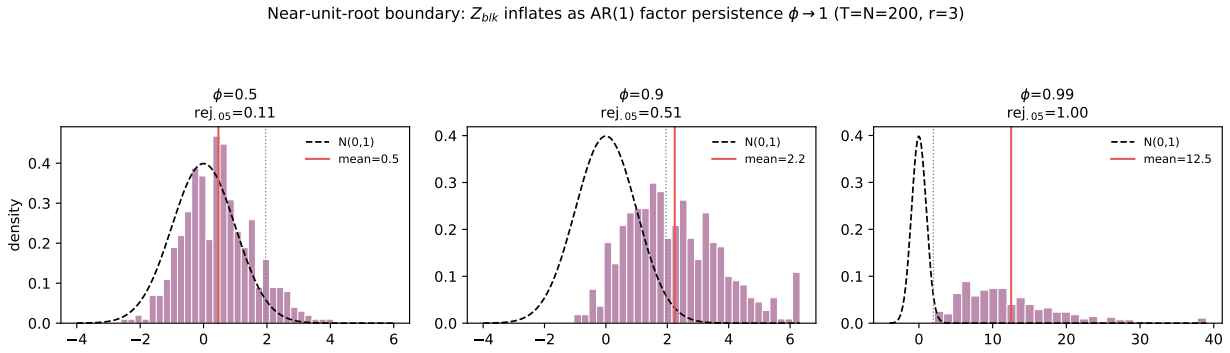


Figure 9: Near-unit-root boundary: Z_{blk} inflates as the AR(1) factor persistence $\phi \to 1$ ($T = N = 200$, $r = 3$).

J Quantile factor model application

J.1 Quantile Factor Model

We next illustrate factor dipping as a stress test for PC-based conclusions and as a visualization tool for PC uncertainty. Chen et al. (2021) introduce quantile factor models (QFM), which allow latent factors to influence the entire conditional distribution of observables (beyond the mean), and propose quantile factor analysis (QFA) via quantile regression for consistent estimation at each quantile τ.

Chen et al. (2021) compare QFA factors $\hat{F}^{\tau}_{QFA}$ (with quantile-specific rank $\hat{r}_{rank}$) to the 8 standard PC factors $\hat{F}_{PCA}$ extracted from FRED-QD data using Bai and Ng (2002),

regressing each column of $\hat{F}^{\tau}_{QFA}$ on $\hat{F}_{PCA}$ and reporting R^2 values (see Table III of Chen et al. (2021), also reproduced in the top panel of Table 13). They conclude that QFA factors are well explained by PC factors near the median and upper quantiles ($\tau \approx 0.50, 0.75$), where R^2 values approach 1.

To stress-test this interpretation, we apply factor dipping with the two strongest transformations in the library, $A_1 = \mathbf{A}_T(0.6)$ and $A_2 = \mathbf{A}_T(-0.6)$ (both $\|A\| \approx 4$, dominated by $\|F\Lambda'\| \approx 100$), yielding dipped PC estimates $\hat{F}_{PCA,1}$ and $\hat{F}_{PCA,2}$. Under the strong-factor assumption, regressions of $\hat{F}^{\tau}_{QFA}$ on these dipped estimates should yield similar adjusted R^2 patterns. The middle and bottom panels of Table 13 show otherwise. The leading QFA factor remains well explained, but the higher factors no longer do: at $\tau = 0.50$ the fits that are near unity under the original PC estimates (0.96, 0.95, 0.92 for factors 3–5) fall to 0.54 for the second factor under $\mathbf{A}_T(0.6)$ and to 0.45, 0.23, 0.25 for factors 3–5 under $\mathbf{A}_T(-0.6)$; the pattern repeats at $\tau = 0.75$. This indicates that uncertainty in PC estimates, potentially induced by weak factors, undermines their claim of close alignments between PCA and QFA at these τ. The effect is not an artefact of the exponential-smoothing family: across the full 65-dip library the adjusted R^2 of the higher QFA factors (3–5) falls to an envelope minimum of 0.45, 0.23 and 0.25 at $\tau = 0.50$ (and 0.39, 0.37, 0.20 at $\tau = 0.75$)—driven independently by autoregressive and moving-average dips as well as KMS—while the leading factor stays above 0.89 throughout. The two panels of Table 13 display the two strongest KMS members as representative of this envelope.

K FAVAR: subspace preservation and impulse-response sensitivity

This appendix supports the FAVAR discussion of Section 7. Two facts organise it: the estimated factor *space* is nearly invariant to the representation, while the *impulse responses* built on it range from fully robust to fully fragile depending on the variable.

The factor space is preserved. Table 14 reports, for each retained PC factor, the minimum canonical correlation between the original and dipped factor spaces over all 65 dips of the recommended library. The first five factors are preserved to canonical correlation at least 0.99; only the seventh and eighth—at the edge of the retained set—move appreciably. Dipping therefore does not contest the Bernanke et al. (2005) factor count: whatever number of factors one retains up to about five, the span is the same across every admissible representation. The representation-sensitivity documented in the main text is thus not a statement

Table 13: MACRO FORECASTING: COMPARISON OF $\hat{F}^{\tau}_{QFA}$ and $\hat{F}_{PCA}$. This table contains three sections, which all present R^2 of regressing each column of $\hat{F}^{\tau}_{QFA}$ on the 8 PC factors $\hat{F}_{PCA}$. These three sections differ in their PCA set up, where the first is the classical PCA applied by Chen et al. (2021), the second and third section uses different Toeplitz matrices to dip the factors for alternative estimates $\hat{F}_{PCA,1}$ and $\hat{F}_{PCA,2}$.

		Elements of $\hat{F}^{\tau}_{QFA}$ on $\hat{F}_{PCA}$					
τ	$\hat{r}_{rank}$	1	2	3	4	5	6
0.01	1	0.657					
0.05	2	0.733					
0.10	2	0.796	0.871				
0.25	4	0.952	0.932	0.939	0.890		
0.50	5	0.993	0.976	0.964	0.945	0.923	
0.75	5	0.906	0.945	0.943	0.903	0.882	
0.90	2	0.316	0.911				
0.95	1	0.261					
0.99	1	0.266					
		Elements of $\hat{F}^{\tau}_{QFA}$ on $\hat{F}_{PCA,1} = \mathbf{A}_T(0.6)$-dipped					
τ	$\hat{r}_{rank}$	1	2	3	4	5	6
0.01	1	0.674					
0.05	2	0.731					
0.10	2	0.740	0.757				
0.25	4	0.934	0.820	0.904	0.697		
0.50	5	0.988	0.538	0.939	0.743	0.737	
0.75	5	0.863	0.897	0.945	0.820	0.742	
0.90	2	0.223	0.890				
0.95	1	0.196					
0.99	1	0.183					
		Elements of $\hat{F}^{\tau}_{QFA}$ on $\hat{F}_{PCA,2} = \mathbf{A}_T(-0.6)$-dipped					
τ	$\hat{r}_{rank}$	1	2	3	4	5	6
0.01	1	0.574					
0.05	2	0.618					
0.10	2	0.682	0.736				
0.25	4	0.854	0.817	0.467	0.648		
0.50	5	0.893	0.888	0.446	0.228	0.253	
0.75	5	0.821	0.795	0.393	0.374	0.198	
0.90	2	0.197	0.810				
0.95	1	0.151					
0.99	1	0.130					

about the factor space but about how the two-step FAVAR maps that robust space into structural impulse responses.

Table 14: Minimum canonical correlation between the original and dipped PC factor spaces over the full 65-dip library, by factor rank, on the Bernanke et al. (2005) FAVAR panel. The space is preserved (canonical correlation at least 0.99) through five factors and degrades only at ranks seven and eight.

factor rank	1	2	3	4	5	6	7	8
min. canon. corr.	1.00	1.00	1.00	1.00	0.99	0.97	0.89	0.27

The impulse responses, grouped. Inspecting the response of every one of the 116 observed variables—the full set of per-variable panels, each showing $k = 1, 3, 5$ with the original estimate, the bootstrap band, and the dipped fan, is provided in the replication materials—yields the grouping summarizedin Table 15. The gradient is systematic: the policy instrument, exchange rates, and several price deflators are representation-robust; the long end of the term structure and industrial production keep their sign and broad shape while the dips spread their magnitude; and money and credit aggregates, the equity market, the labour market, and the headline price indices are representation-fragile, their dipped fans wide enough to unsettle the sign or shape of the response. In every fragile group the fan widens materially from $k = 3$ to $k = 5$, which is the sense in which the fifth factor trades representation-robustness for a negligible change in the point estimate.

Table 15: Representation-robustness of the FAVAR impulse responses by FRED-MD variable group, read from the full per-variable dipped responses across the 65-dip library. "Robust": dipped fan tight, sign and shape stable across k. "Shape robust": sign and profile stable, magnitude and persistence spread. "Fragile": dipped fan wide enough to change the sign or qualitative shape, worsening from $k = 3$ to $k = 5$.

robustness	variable groups	examples
robust	policy rate, exchange rates, some deflators	FEDFUNDS, EXUSUKx, PCE/WPS deflators
shape robust	long-end term structure, industrial production	GS5, GS10, INDPRO
fragile	money & credit, equities, labour, headline prices	M2SL, S&P 500, UNRATE, CPIAUCSL

L Strong-factor preservation across the empirical panels

Table 16 validates, on all three empirical panels of Section 7, that the recommended library leaves the well-separated factors intact and perturbs only the near-degenerate ones, as Theorem 2.1 requires.

Table 16: Strong-factor preservation across the full recommended library (65 dips). For each empirical panel and each PC factor (ordered by eigenvalue), the eigenvalue gap $n_j = \lambda_j/\lambda_{j+1}$ and the *minimum* canonical correlation between the original and dipped PC factor over all 65 dips. Well-separated factors ($n_j > 1.5$) are preserved by every dip; only near-degenerate factors ($n_j \approx 1$) are perturbed.

panel		PC factor (by eigenvalue rank)							
		1	2	3	4	5	6	7	8
Stock–Watson	gap n_j	5.3	1.6	1.4	1.9	1.5	1.3	1.4	1.4
(FRED-MD)	min CC	1.00	1.00	1.00	1.00	1.00	0.98	0.90	0.34
Three-pass	gap n_j	19.2	1.7	1.8	2.2	1.1	1.1	1.4	–
(returns)	min CC	1.00	1.00	1.00	1.00	0.97	0.89	0.48	–
Quantile FM	gap n_j	2.5	1.1	1.8	1.1	1.3	1.1	1.1	1.1
(FRED-QD)	min CC	0.99	0.99	0.96	0.72	0.60	0.23	0.17	0.00

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